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QuantileGS: Quantile-Based Conformal Bounds on RGB Rendering Errors for Novel View Synthesis with 3D Gaussian Splatting

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Abstract

Uncertainty quantification for novel view synthesis with 3D Gaussian Splatting (3DGS) assesses rendering reliability at unseen viewpoints. This reliability depends on the discrepancy between rendered and true colors, which cannot be measured without ground-truth images. We therefore quantify uncertainty in this unknown discrepancy through upper bounds on per-pixel RGB mean absolute error. However, learned bounds may underestimate these errors and overstate rendering reliability, motivating calibration to a prescribed coverage level. In this paper, we propose QuantileGS, a post-hoc framework that combines learned rendering-error quantiles with scale-normalized conformal calibration to produce spatially adaptive error bounds. The framework comprises three phases. Phase 1, Rendering-Error Computation, measures RGB mean absolute error between observed images and predictions from a frozen 3DGS model. Phase 2, Ordered Gaussian Quantile Fields, learns two scalar attributes per Gaussian to predict lower and upper rendering-error quantiles. Phase 3, Scale-Normalized Conformal Calibration, calibrates the upper prediction on held-out images, using the quantile gap to scale the correction. Under the stated assumptions, the resulting bounds provide finite-sample marginal coverage. At a nominal coverage level of 90%*, calibration raises mean empirical test coverage from* 69.6% *to* 89.6% *on LLFF and from* 80.1% *to* 92.1% *on Mip-NeRF 360. On LF, coverage remains conservative at* 96.3% *before and* 96.4% *after calibration. Applications to active view selection, Gaussian pruning, and RGB-D simultaneous localization and mapping illustrate how these bounds inform downstream decisions.*

1. Introduction

Uncertainty quantification (UQ) for novel view synthesis (NVS) with 3D Gaussian Splatting (3DGS) [7, 25] assesses the reliability of images rendered from previously unseen viewpoints. Despite advances in generative view synthesis [10], efficient rendering [46], and sparse-view reconstruction [49, 51, 52], high reconstruction quality on observed views does not ensure accurate predictions at unseen viewpoints. Occlusion, limited observations, and view-dependent appearance leave scene properties weakly constrained, producing errors that vary across pixels and viewpoints. The reliability of a rendered pixel depends on its agreement with the ground-truth RGB at the requested viewpoint, which is unavailable at inference. We therefore characterize the predictive uncertainty of a fixed 3DGS model through the RGB errors its predictions may incur. Specifically, we estimate upper quantiles of per-pixel RGB mean absolute error (RGB-MAE), the average absolute discrepancy across the three color channels. This expresses uncertainty in terms of how large the unknown rendering error may be.

Existing methods estimate rendering uncertainty through parameter variation [22, 28, 30, 40], rendering sensitivity [23], and cross-view consistency [34]. Others use reconstruction residuals and observation coverage to predict errors or identify unreliable regions [14, 17, 29]. These approaches characterize different aspects of rendering reliability. Prediction variability and rendering sensitivity describe how outputs change, but do not by themselves measure agreement with ground truth. For example, predictions from different model samples can agree closely while sharing a systematic color bias. Learning from reconstruction residuals connects uncertainty more directly to observed RGB discrepancies, providing a basis for interpreting the reliability of a fixed renderer in terms of its prediction errors.

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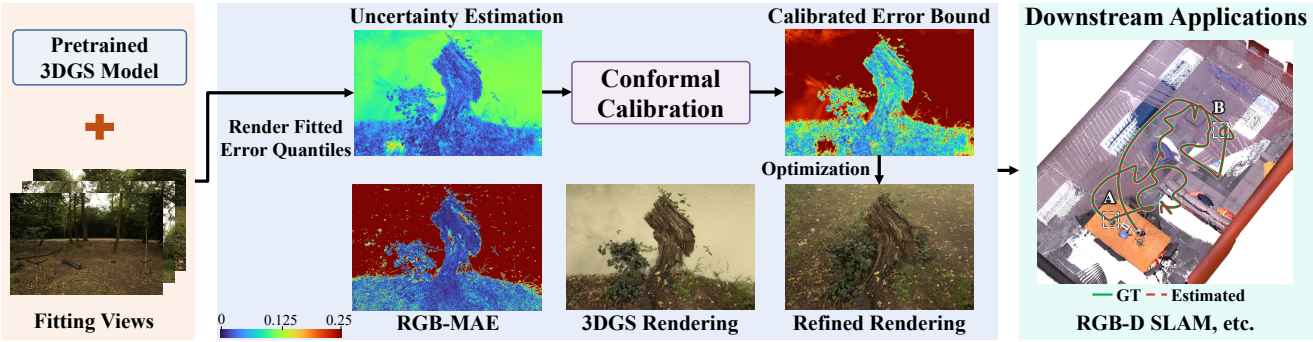

Figure 1. QuantileGS estimates rendering-error uncertainty and uses conformal calibration to produce pixel-wise RGB-MAE upper bounds. The reference error map compares the original 3DGS rendering with ground-truth RGB. Calibrated UQ guides model optimization to produce the refined rendering. We apply the resulting bounds to RGB-D SLAM, as well as active view selection and Gaussian pruning.

However, these methods do not generally provide RGB-MAE upper bounds at a prescribed reliability level. Learned upper error quantiles characterize large deviations that mean-error estimates may miss, but need not attain their nominal levels on unseen observations, motivating calibration with held-out errors [5, 6]. Coverage is the probability that a new pixel's RGB-MAE does not exceed its predicted bound, with tightness assessed at comparable coverage. Whereas recent work provides view-level guarantees for RGB prediction boxes [11], we target spatially adaptive scalar RGB-MAE bounds with marginal coverage.

To this end, we propose QuantileGS, a post-hoc framework that estimates and calibrates rendering-error uncertainty with a frozen pretrained 3DGS model. Two non-negative attributes per Gaussian parameterize ordered lower and upper RGB-MAE quantile fields. Shared across views, these attributes yield view-dependent predictions through the original α-blending weights and fixed background values. Conformal calibration adjusts the upper prediction using the floored interquantile gap as a local scale [5, 33]. Calibration and test views are excluded from RGB pretraining and quantile fitting. Novel-view bounds require camera parameters but no target RGB. Under exchangeability of pixel-level calibration and test scores, conditional on fitting, the bounds provide finite-sample marginal coverage [26]. This guarantee averages over the calibration sample and a new pixel observation, without implying a separate coverage guarantee for each view or pixel location.

Experiments on LF, LLFF, and Mip-NeRF 360 assess error ranking, empirical coverage, and mean upper bounds. Prior work uses uncertainty for diffusion-based scene refinement [9], active view selection (AVS) [27], Gaussian pruning [19], and camera tracking [37]. Gaussian representations also support simultaneous localization and mapping (SLAM) [24, 41, 45] and dynamic driving-scene reconstruction [43]. We assess downstream utility by using the bounds to weight prediction-variance reduction for AVS, estimate Gaussian deletion costs, and control reference RGB corrections and photometric residual scales for RGB-D SLAM. Fig. 1 illustrates the framework and these applications, including renderings before and after calibrated UQ-guided optimization. Calibration preserves the original RGB rendering. Its coverage guarantee applies to the fixed renderer, while downstream benefits are evaluated empirically.

Our main contributions are summarized as follows:

- We introduce ordered Gaussian error-quantile fields that learn two attributes per Gaussian while preserving a pretrained 3DGS renderer.
- We develop scale-normalized conformal calibration for spatially adaptive RGB-MAE upper bounds with finite-sample marginal coverage under conditional score exchangeability.
- We evaluate uncertainty estimation and calibration on LF, LLFF, and Mip-NeRF 360, and examine their utility in active view selection, Gaussian pruning, and RGB-D SLAM.

2. Related Work

2.1. Uncertainty Quantification for View Synthesis

NVS uncertainty estimates draw on probabilistic modeling, rendering sensitivity, and multi-view evidence. S-NeRF [35] and CF-NeRF [36] model radiance distributions through variational inference and normalizing flows, respectively. Bayes' Rays [16] uses a post-hoc Laplace approximation. For 3DGS, approaches include stochastic Gaussian attributes [2], variational multi-scale representations [28], learned parameter manifolds [30], and Horseshoe shrinkage priors [40]. FisherRF [23] instead measures sensitivity through rendering derivatives.

Reconstruction evidence supports error prediction from Gaussian features [17] and uncertainty fitting with a fixed renderer [14]. WarpRF [34] uses cross-view inconsisten-

cies, while OCSplats [29] uses observation completeness to refine noise estimates. Abstract Rendering [21] certifies image variations under specified camera and scene perturbations, rather than errors against observed RGB. QuantileGS learns ordered rendering-error quantiles and calibrates their upper prediction while preserving the RGB model.

2.2. Conformal Prediction and Error Calibration

Split conformal prediction provides finite-sample marginal coverage under exchangeability [26]. Residual normalization [26] and conformalized quantile regression [33] adapt interval widths, while score optimization targets efficiency [42]. Conformal risk control extends calibration to expected bounded monotone losses [4], with applications to confidence masks for generative super-resolution [1]. Further work studies guarantees for specified subgroups and covariate shifts [15], and calibration under distribution shift using optimal transport and unlabeled target data [12].

For NVS, Meta-Calibrator [3] learns scene-specific calibration curves across scenes. VSCP [11] constructs RGB prediction boxes that cover a prescribed fraction of pixels with a specified probability under view-level exchangeability. QuantileGS calibrates a scalar RGB-MAE bound using the predicted interquantile gap for spatial normalization. Its guarantee is marginal under exchangeability of pixel-level calibration and test scores, rather than conditional at every pixel or view.

2.3. Downstream Applications of Radiance Fields

View acquisition methods use predicted uncertainty reduction [32], information-based criteria [23, 39], or anisotropic visibility [44]. Gaussian compression uses global significance [13], reconstruction-error sensitivity [19], or pruning integrated with efficient rasterization [18].

Gaussian representations also support RGB-D tracking and mapping [24, 48]. SEGS-SLAM [38] incorporates structural cues and appearance embeddings, while WildGS-SLAM [50] and VarSplat [37] incorporate learned uncertainty into tracking. QuantileGS uses calibrated error bounds to weight prediction-variance reduction for acquisition, rank pruning risk, and scale photometric tracking residuals. Downstream gains are evaluated empirically.

3. Preliminaries

3.1. 3D Gaussian Splatting

3DGS [25] represents a scene by Gaussian primitives $\mathcal{G}$, with pretrained parameters $\boldsymbol{\theta}$ encoding their centers, scales, rotations, opacities, and view-dependent color coefficients.

For a camera ray $\boldsymbol{x}$, let $\alpha_j(\boldsymbol{x}) \in [0,1]$ be the effective opacity of Gaussian j, incorporating its opacity and projected footprint. Front-to-back compositing uses

$$\begin{aligned} T_j(\boldsymbol{x}) &= \prod_{\ell \prec_{\boldsymbol{x}} j} \big(1-\alpha_\ell(\boldsymbol{x})\big), \\ w_j(\boldsymbol{x}) &= T_j(\boldsymbol{x})\alpha_j(\boldsymbol{x}), \end{aligned} \tag{1}$$

where $\ell \prec_{\boldsymbol{x}} j$ denotes depth ordering along the ray, T_j is transmittance, and w_j is the blending weight. The accumulated opacity and residual transmittance are

$$A(\boldsymbol{x}) = \sum_{j\in\mathcal{G}} w_j(\boldsymbol{x}), \qquad r(\boldsymbol{x}) = 1 - A(\boldsymbol{x}). \tag{2}$$

The Gaussian and background weights are therefore nonnegative and sum to one.

For Gaussian colors $\boldsymbol{c}_j(\boldsymbol{d})$ at viewing direction $\boldsymbol{d}$ and a fixed RGB background $\boldsymbol{b} \in [0,1]^3$, the composited color is

$$\boldsymbol{C}(\boldsymbol{x}) = \sum_{j\in\mathcal{G}} w_j(\boldsymbol{x})\boldsymbol{c}_j(\boldsymbol{d}) + r(\boldsymbol{x})\boldsymbol{b}. \tag{3}$$

The RGB prediction at pixel i is

$$\hat{\boldsymbol{y}}_i = f_{\boldsymbol{\theta}}(\boldsymbol{x}_i) = \mathrm{clip}_{[0,1]^3}\big(\boldsymbol{C}(\boldsymbol{x}_i)\big).$$

Freezing $\boldsymbol{\theta}$ fixes the weights for each ray. QuantileGS reuses them to composite learned Gaussian error attributes with fixed scalar background values.

3.2. Problem Formulation

Let X be a random pixel ray and $\boldsymbol{Y} \in [0,1]^3$ its ground-truth RGB. For the fixed renderer, the RGB-MAE is $E = \|f_{\boldsymbol{\theta}}(X) - \boldsymbol{Y}\|_1/3 \in [0,1]$, with observed values e_i defined in Eq. (6). Its conditional quantile at level $\nu \in (0,1)$ is

$$q_\nu(\boldsymbol{x}) = \inf\big\{t : \mathbb{P}(E \le t \mid X = \boldsymbol{x}) \ge \nu\big\}. \tag{4}$$

For a novel view without target RGB, we seek a pixel-wise upper error bound $\hat{u}_\tau$ satisfying

$$\mathbb{P}\{E_{\mathrm{new}} \le \hat{u}_\tau(X_{\mathrm{new}})\} \ge \tau, \tag{5}$$

where $\tau \in (0,1)$ is the target coverage. The probability is marginal over the calibration sample and a new pixel observation, with fitting information held fixed.

4. QuantileGS

QuantileGS constructs calibrated pixel-wise RGB-MAE bounds through the three phases in Fig. 2. We partition views into disjoint *fitting views* $\mathcal{V}_{\mathrm{fit}}$, *calibration views* $\mathcal{V}_{\mathrm{cal}}$, and *test views* $\mathcal{V}_{\mathrm{test}}$, with corresponding pixel-index sets $\mathcal{I}_{\mathrm{fit}}$, $\mathcal{I}_{\mathrm{cal}}$, and $\mathcal{I}_{\mathrm{test}}$. We use $\mathcal{V}_{\mathrm{fit}}$ for RGB pretraining and quantile fitting, $\mathcal{V}_{\mathrm{cal}}$ only for calibration, and $\mathcal{V}_{\mathrm{test}}$ only for evaluation.

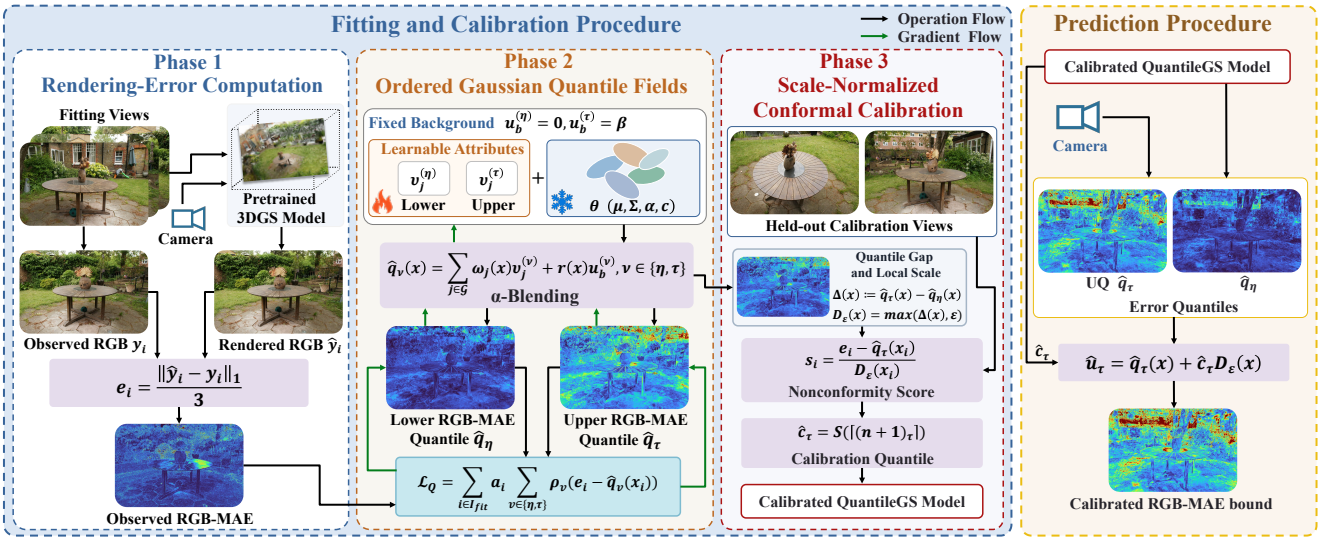

Figure 2. **QuantileGS overview.** Phase 1 computes RGB-MAE on $\mathcal{V}_{\text{fit}}$ using a frozen pretrained 3DGS model. Phase 2 learns ordered Gaussian error quantiles with fixed blending weights and background contributions. Phase 3 calibrates the upper prediction on $\mathcal{V}_{\text{cal}}$, using the interquantile gap with a positive floor as the local scale. The prediction procedure renders the learned quantiles for a new view and applies a locally scaled conformal correction to produce an RGB-MAE upper-bound map.

4.1. Rendering-Error Computation

We start from a 3DGS model pretrained on $\mathcal{V}_{\text{fit}}$ [25]. All original parameters $\boldsymbol{\theta}$ remain frozen throughout QuantileGS. For a pixel ray $\boldsymbol{x}_i$ specified by its camera parameters, let $\boldsymbol{y}_i \in [0,1]^3$ be the observed RGB and $\hat{\boldsymbol{y}}_i = f_{\boldsymbol{\theta}}(\boldsymbol{x}_i) \in [0,1]^3$ the clipped rendering. The fixed fitting targets are

$$e_i = \frac{\|\hat{\boldsymbol{y}}_i - \boldsymbol{y}_i\|_1}{3} \in [0,1], i \in \mathcal{I}_{\text{fit}}. \tag{6}$$

4.2. Ordered Gaussian Quantile Fields

For each Gaussian $j \in \mathcal{G}$, we learn attributes $v_j^{(\eta)}$ and $v_j^{(\tau)}$ for lower and upper error quantiles at levels $0 < \eta < \tau < 1$. We use $\eta = 0.5$, and τ is the target coverage level.

For the RGB background $\boldsymbol{b} \in [0,1]^3$, define

$$d_i = \frac{\|\boldsymbol{y}_i - \boldsymbol{b}\|_1}{3}, i \in \mathcal{I}_{\text{fit}}. \tag{7}$$

Sort these $N_{\text{fit}} = |\mathcal{I}_{\text{fit}}|$ values as $d_{(1)} \le \cdots \le d_{(N_{\text{fit}})}$ and set

$$\beta = d_{(\lceil N_{\text{fit}} \tau \rceil)}, \quad u_b^{(\eta)} = 0, u_b^{(\tau)} = \beta. \tag{8}$$

Thus, β is an empirical upper quantile of the observed-color-to-background error. We compute it once from fitting colors and store it as a fixed buffer without gradients.

Let $w_j(\boldsymbol{x}) \ge 0$ be the frozen α-blending weights and $r(\boldsymbol{x}) = 1 - \sum_j w_j(\boldsymbol{x}) \ge 0$ the residual transmittance. For $\nu \in \{\eta, \tau\}$,

$$\hat{q}_\nu(\boldsymbol{x}) = \sum_{j \in \mathcal{G}} w_j(\boldsymbol{x}) v_j^{(\nu)} + r(\boldsymbol{x}) u_b^{(\nu)}. \tag{9}$$

Phase 2 of Fig. 2 composites Gaussian-specific scalar attributes into pixel-wise error-quantile predictions. The index j is summed out in $\hat{q}_\nu(\boldsymbol{x})$, as in the RGB rendering $\boldsymbol{C}(\boldsymbol{x})$, while each $v_j^{(\nu)}$ remains specific to Gaussian j. Unlike the directional colors $\boldsymbol{c}_j(\boldsymbol{d})$, these scalar attributes are shared across views, with view dependence entering through the blending weights $w_j(\boldsymbol{x})$ and residual transmittance $r(\boldsymbol{x})$. The attributes are learned jointly from pixel-wise RGB-MAE targets, without requiring separate error targets for individual Gaussians.

The lower field contains only Gaussian contributions, and the upper field includes the fixed offset $r(\boldsymbol{x})\beta$. We impose the ordering constraints

$$v_j^{(\eta)} \ge 0, v_j^{(\tau)} - v_j^{(\eta)} \ge 0, j \in \mathcal{G}, \tag{10}$$

using $v_j^{(\eta)} = \ell_j$ and $v_j^{(\tau)} = \ell_j + h_j$ with $\ell_j, h_j \ge 0$. Since $\beta \ge 0$, the composited gap satisfies

$$\begin{aligned} \Delta(\boldsymbol{x}) &:= \hat{q}_\tau(\boldsymbol{x}) - \hat{q}_\eta(\boldsymbol{x}) \\ &= \sum_{j \in \mathcal{G}} w_j(\boldsymbol{x}) h_j + r(\boldsymbol{x})\beta \ge 0. \end{aligned} \tag{11}$$

The fields cannot cross, but their gap can vanish, motivating the positive floor in the calibration scale D_ε below.

With the pinball loss

$$\rho_\nu(z) = \max\{\nu z, (\nu - 1)z\}, \tag{12}$$

we jointly fit both fields by minimizing

$$\mathcal{L}_{\text{UQ}} = \sum_{i \in \mathcal{I}_{\text{fit}}} a_i \sum_{\nu \in \{\eta,\tau\}} \rho_\nu(e_i - \hat{q}_\nu(\boldsymbol{x}_i)), \tag{13}$$

where $a_i \geq 0$ are fixed weights summing to one. We project ℓ_j, h_j onto $[0, \infty)$ after each first-order update. Gradients pass through the fixed blending operation to these attributes; the RGB parameters $\boldsymbol{\theta}$, weights w_j, and background statistic β remain fixed.

To establish convexity in attribute space, stack the Gaussian attributes and pixel-i weights into $\boldsymbol{v}^{(\nu)}$ and $\boldsymbol{w}_i$. With fixed offsets $b_{i\eta} = 0$ and $b_{i\tau} = r(\boldsymbol{x}_i)\beta$, the predictions are affine: $\hat{q}_\nu(\boldsymbol{x}_i) = \boldsymbol{w}_i^\top \boldsymbol{v}^{(\nu)} + b_{i\nu}$. Each loss term is a maximum of affine functions, so Eq. (13) is convex under the linear ordering constraints. Introducing auxiliary variables $\xi_{i\nu}$ gives the equivalent linear program

$$\begin{aligned} \min_{\substack{\boldsymbol{v}^{(\eta)}, \boldsymbol{v}^{(\tau)}, \\ \boldsymbol{\xi}}} \quad & \sum_{i \in \mathcal{I}_{\text{fit}}} \sum_{\nu \in \{\eta, \tau\}} a_i \xi_{i\nu} \\ \text{s.t.} \quad & \xi_{i\nu} \geq \nu(e_i - \boldsymbol{w}_i^\top \boldsymbol{v}^{(\nu)} - b_{i\nu}), \\ & \xi_{i\nu} \geq (\nu - 1)(e_i - \boldsymbol{w}_i^\top \boldsymbol{v}^{(\nu)} - b_{i\nu}), \\ & \boldsymbol{v}^{(\eta)} \geq \mathbf{0}, \\ & \boldsymbol{v}^{(\tau)} - \boldsymbol{v}^{(\eta)} \geq \mathbf{0}. \end{aligned} \tag{14}$$

The first two constraints apply to every fitting pixel and both quantile levels, with vector inequalities taken componentwise. Background offsets are fixed rather than optimized. In practice, we optimize the pinball objective directly.

4.3. Scale-Normalized Conformal Calibration

We freeze the fitted fields and use $n = |\mathcal{I}_{\text{cal}}|$ pixel observations from $\mathcal{V}_{\text{cal}}$ to calibrate the upper prediction. Fix $\varepsilon > 0$ and τ before calibration, and define

$$D_\varepsilon(\boldsymbol{x}) = \max\{\Delta(\boldsymbol{x}), \varepsilon\}. \tag{15}$$

The interquantile gap measures the predicted spread between the median and the target upper quantile. It allows the calibration correction to vary spatially, while the positive floor prevents division by a vanishing gap.

Reindexing the calibration pixels by $i = 1, \ldots, n$, compute their RGB-MAE and normalized scores

$$s_i = \frac{e_i - \hat{q}_\tau(\boldsymbol{x}_i)}{D_\varepsilon(\boldsymbol{x}_i)}. \tag{16}$$

Positive scores indicate that the observed error exceeds the upper prediction, whereas negative scores indicate that it falls below it. Normalization expresses these deviations relative to the floored local scale, rather than their absolute RGB-MAE magnitude.

Sort the scores as $s_{(1)} \leq \cdots \leq s_{(n)}$, append $s_{(n+1)} = +\infty$, and set

$$\hat{c}_\tau = s_{(k_\tau)}, k_\tau = \lceil (n+1)\tau \rceil. \tag{17}$$

The scalar correction adapts to each ray through the same scale:

$$\hat{u}_\tau(\boldsymbol{x}) = \hat{q}_\tau(\boldsymbol{x}) + \hat{c}_\tau D_\varepsilon(\boldsymbol{x}). \tag{18}$$

Phase 3 of Fig. 2 illustrates how a single calibration coefficient produces spatially varying corrections through $D_\varepsilon(\boldsymbol{x})$. A positive $\hat{c}_\tau$ increases the upper predictions, whereas a negative value reduces them. Both the fitted quantile fields and the pretrained RGB renderer remain frozen. Calibration therefore changes the reported error bounds without changing the RGB predictions or their realized errors. Let $\mathcal{F}_{\text{fit}}$ contain all fitting information, including RGB pretraining and optimization randomness. The following theorem explains the gap normalization.

Theorem 1 (Joint local-scale calibration)**.** *Condition on $\mathcal{F}_{\text{fit}}$. For the n calibration observations and one new test observation, suppose $E_i = \mu(X_i) + \sigma(X_i)Z_i$ and $\hat{q}_\nu(\boldsymbol{x}) = \mu(\boldsymbol{x}) + \sigma(\boldsymbol{x})\hat{z}_\nu$, $\nu \in \{\eta, \tau\}$, with fixed functions μ, σ, $\sigma > 0$, and finite fitting-dependent constants $\hat{z}_\eta < \hat{z}_\tau$. Assume the standardized residuals are i.i.d. and their vector is independent of the entire ray vector. If $\Delta(X_i) \geq \varepsilon$ almost surely for $i = 1, \ldots, n+1$, the normalized scores are conditionally i.i.d. and*

$$\mathbb{P}\{E_{\text{new}} \leq \hat{u}_\tau(X_{\text{new}}) \mid \mathcal{F}_{\text{fit}}\} \geq \tau. \tag{19}$$

The proof is in Appendix A. Coverage is marginal, and view separation alone does not establish the assumptions. For an active floor, exchangeability of the actual floored scores is sufficient for the same guarantee. This marginal guarantee does not imply simultaneous coverage of all pixels in a view or a separate coverage guarantee for each view.

4.4. Prediction Procedure

After calibration, the RGB model, Gaussian error attributes, background values, and calibration parameters remain fixed. Given a new camera pose and intrinsics, we render the RGB image with $f_{\boldsymbol{\theta}}$ and the two quantile fields using Eq. (9). Their floored gap, defined in Eq. (15), scales the correction in Eq. (18), yielding $\hat{u}_\tau(\boldsymbol{x})$. We return

$$u(\boldsymbol{x}) = \text{clip}_{[0,1]}\big(\hat{u}_\tau(\boldsymbol{x})\big). \tag{20}$$

Clipping preserves the coverage guarantee because RGB-MAE lies in $[0, 1]$. Prediction requires no target RGB and returns the unchanged RGB rendering together with a spatially aligned RGB-MAE upper-bound map.

4.5. Downstream Applications

We use calibrated RGB-MAE bounds $u(\boldsymbol{x})$ to guide three downstream applications. For *active view selection*, the bounds weight the reduction of a local prediction-variance proxy, favoring views informative about regions with larger bounds. For *Gaussian pruning*, they combine with estimated deletion perturbations to assess deletion costs while

accounting for unknown rendering residuals. For *RGB-D SLAM*, they limit reference-color corrections toward observed fitting RGB and set robust scales for photometric residuals during pose estimation. The coverage guarantee applies to rendering errors of the fixed model, while downstream benefits are evaluated empirically. Application-specific objectives, observation splits, and implementation details are provided in Appendix B.

5. Experiments

5.1. Uncertainty Quantification in NVS

Datasets and protocol. We evaluate on LF, LLFF, and Mip-NeRF 360. LF [47] contains four scenes: africa, basket, statue, and torch. Following [28], we retain 4–5 reconstruction views and four test views per scene. We select eight additional views using camera poses and split them into four for fitting directional correction and four for gap-based calibration, none of which are used for reconstruction or testing. LLFF [31] contains eight forward-facing scenes, with every eighth image held out for testing and images downscaled by $4\times$. Mip-NeRF 360 [8] contains nine scenes. Following [25], we hold out every eighth image and downscale indoor and outdoor images by $2\times$ and $4\times$, respectively. QuantileGS uses disjoint fitting and calibration sets, with the underlying 3DGS model frozen throughout UQ fitting and calibration. All experiments use an NVIDIA RTX 3090 with 48 GB memory.

Baselines and metrics. We compare against Variational-3DGS [28], FisherRF [23], PRIMU [17], and Manifold [30], using their official implementations. We report PSNR, SSIM, and LPIPS for rendering, and MAE-based AUSE [20] and Pearson correlation with RGB-MAE for uncertainty quality. QuantileGS-U and QuantileGS-C denote predictions before and after calibration and share the same frozen RGB renderer. For each scene, let $N_{\text{test}} = |\mathcal{I}_{\text{test}}|$. For QuantileGS-C, empirical coverage and mean bound are

$$\begin{aligned} \widehat{\text{Cov}}_\tau &= \frac{1}{N_{\text{test}}} \sum_{i \in \mathcal{I}_{\text{test}}} \mathbb{1}\{e_i \le u(\boldsymbol{x}_i)\}, \\ \overline{u} &= \frac{1}{N_{\text{test}}} \sum_{i \in \mathcal{I}_{\text{test}}} u(\boldsymbol{x}_i), \end{aligned} \tag{21}$$

where e_i is the observed RGB-MAE at test pixel i. For QuantileGS-U, we replace $u(\boldsymbol{x}_i)$ with the unclipped upper quantile $\hat{q}_\tau(\boldsymbol{x}_i)$ in both expressions.

Coverage measures the fraction of test errors contained by their predicted bounds. At comparable coverage, a smaller mean bound indicates tighter uncertainty estimates. AUSE assesses error ranking, whereas Pearson correlation measures linear association with RGB-MAE. Both coverage and mean bound pool all test pixels within each scene, and dataset-level results average scenes equally.

Results. Tables 1 show that QuantileGS achieves lower mean AUSE and higher Pearson correlation than all baselines on all three datasets. Calibration leaves these metrics broadly unchanged. Fig. 3 compares the spatial correspondence between uncertainty estimates and rendering errors. Calibration raises mean coverage from 69.6% to 89.6% on LLFF and from 80.1% to 92.1% on Mip-NeRF 360, while AUSE and correlation change little. The 96.4% coverage on LF indicates conservative bounds in this setting. More experiments are provided in Appendix C.

5.2. Downstream Applications

5.2.1. Active View Selection

Setting and baselines. We evaluate on nine Mip-NeRF 360 scenes with a budget of 20 reconstruction views. Every eighth view is held out for testing. Indoor and outdoor images are downscaled by $2\times$ and $4\times$, respectively. We compare against Random, FisherRF, Manifold, and PRIMU. All methods use the same four initial views, comprising one fixed view and three views selected sequentially by farthest-point sampling. QuantileGS uses the acquisition rule in Sec. B.1, adding one view every $100n$ optimization steps, where n is the current number of selected views. After each acquisition, 3DGS restarts from the same initialization; the final model is evaluated at global iteration 30,000, without additional reconstruction. RGB training uses random backgrounds, while UQ and evaluation use fixed black backgrounds. Reserved UQ fitting and calibration views are additional observations, excluded from our reconstruction set and candidate pool, and baseline candidate pools remain unchanged. Manifold retains its VI_MC acquisition and 2,000-step per-round training, followed by a separate 30,000-step vanilla 3DGS reconstruction using its selected 20 views. Random results are averaged over seeds 0, 1, and 2. All experiments use an NVIDIA RTX 3090 with 24 GB memory.

Results. Table 2 shows that QuantileGS achieves the highest PSNR, exceeding PRIMU and Manifold by 0.244 and 0.547 dB, respectively. Its SSIM is 0.001 below PRIMU, while LPIPS ties for best at the reported precision. These results compare the complete AVS pipelines under the settings above. Relative to Random, QuantileGS improves PSNR by 1.729 dB and reduces LPIPS by 0.036, supporting the practical utility of the evaluated acquisition pipeline.

5.2.2. Gaussian Pruning

Setting and baselines. We evaluate the strategy in Sec. B.2 on nine Mip-NeRF 360 scenes against PUP 3D-GS [19] and LightGaussian [13], using identical 30k-iteration checkpoints, resolutions, and test views. We reuse the saved calibration splits, containing 21-54 views per scene, excluded from base-model and UQ fitting. QuantileGS uses projected parameters with Adam, $\tau = 0.9$, $\alpha = 0.1$, gap normaliza-

Table 1. Uncertainty quantification, empirical coverage, and novel-view synthesis results on LLFF, Mip-NeRF 360, and LF. The best, second-best, and third-best results for AUSE and Pearson correlation are highlighted, with the best also in bold. Coverage is the mean empirical coverage across scenes, with a nominal coverage level of 90% ($\alpha = 0.1$). It is reported for both QuantileGS variants; — denotes baselines without conformal calibration, for which coverage is not reported.

Dataset	Method	AUSE ↓	Pears. ↑	Coverage (%)	PSNR ↑	SSIM ↑	LPIPS ↓
LLFF	Variational-3DGS	0.381	0.269	—	26.179	0.847	0.143
	FisherRF	0.420	-0.059	—	24.880	0.824	0.156
	PRIMU	0.289	0.364	—	24.508	0.820	0.163
	Manifold	0.436	0.186	—	25.311	0.826	0.151
	QuantileGS-U (ours)	0.250	**0.428**	69.6	24.407	0.796	0.173
	QuantileGS-C (ours)	**0.249**	0.421	89.6			
Mip-NeRF 360	Variational-3DGS	0.514	0.143	—	27.350	0.812	0.225
	FisherRF	0.476	-0.050	—	27.644	0.823	0.206
	PRIMU	0.389	0.214	—	27.686	0.825	0.201
	Manifold	0.444	0.221	—	27.696	0.823	0.207
	QuantileGS-U (ours)	**0.315**	**0.371**	80.1	27.165	0.810	0.214
	QuantileGS-C (ours)	0.316	0.368	92.1			
LF	Variational-3DGS	0.333	0.351	—	18.639	0.673	0.319
	FisherRF	0.542	-0.070	—	26.722	0.899	0.140
	PRIMU	0.321	0.353	—	18.487	0.618	0.350
	Manifold	0.434	0.173	—	25.924	0.877	0.157
	QuantileGS-U (ours)	**0.275**	0.392	96.3	28.967	0.935	0.072
	QuantileGS-C (ours)	**0.275**	**0.393**	96.4			

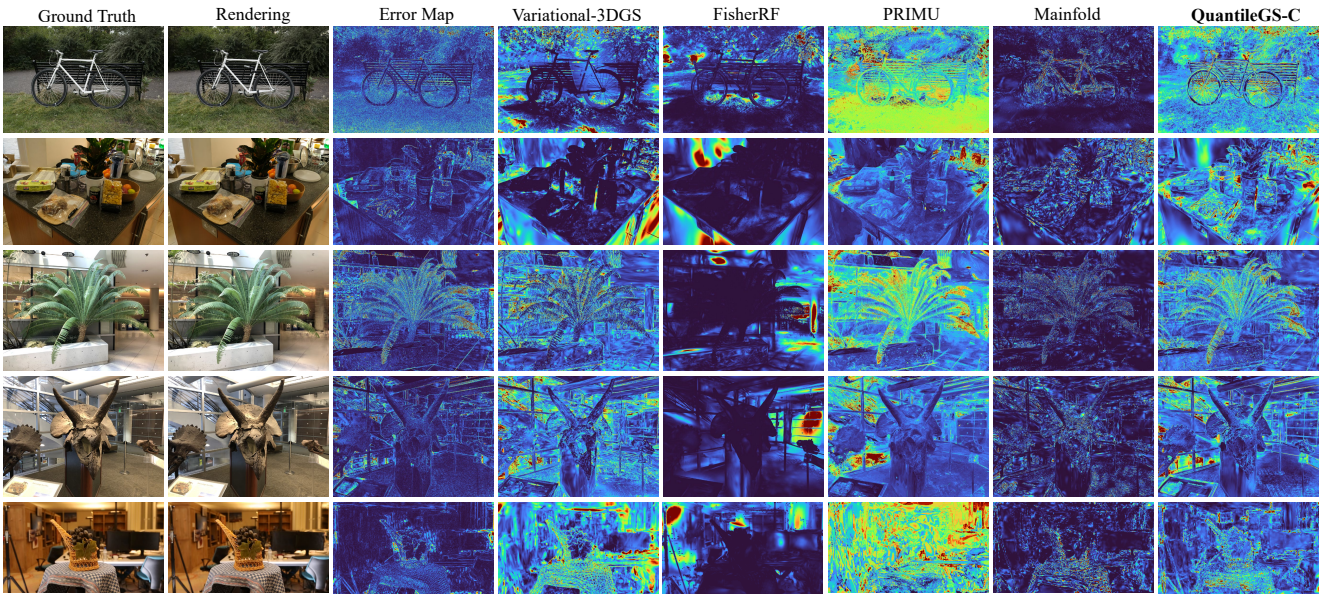

Figure 3. Qualitative comparison of Variational-3DGS [28], FisherRF [23], PRIMU [17], Manifold [30], and QuantileGS-C (Ours). Each scene shows uncertainty maps. Scenes from top to bottom: Mip-NeRF 360 bicycle and counter, LLFF fern and horns, and LF basket.

tion, and two probes per fitting view. All methods retain 20% and then 10% of the original Gaussians. Our method uses two progressive pruning stages per round, with 2.5k refinement updates per stage and no densification. Scoring uses images downscaled by a further factor of four. Refinement uses all non-test training views with UQ frozen. Light-

Table 2. AVS on Mip-NeRF 360: nine-scene means with 20 reconstruction views. The best, second-best, and third-best results are highlighted, with the best also in bold. Ties at the displayed precision share a rank.

Method	PSNR ↑	SSIM ↑	LPIPS ↓
Random	19.361	0.584	0.364
FisherRF	20.316	0.597	0.355
Manifold	20.543	0.609	0.338
PRIMU	20.846	**0.636**	**0.328**
QuantileGS (Ours)	**21.090**	0.635	**0.328**

Table 3. Pruning on Mip-NeRF 360: nine-scene means, seed 0. Within each retention level, the best and second-best results are highlighted, with the best also in bold. Rendering metrics are computed from floating-point images.

Method	Kept	PSNR ↑	SSIM ↑	LPIPS ↓
Unpruned 3DGS	100%	27.170	0.810	0.215
LightGaussian	20%	26.814	0.794	0.257
PUP 3D-GS	20%	27.037	**0.801**	**0.242**
QuantileGS (Ours)	20%	**27.090**	0.784	0.274
LightGaussian	10%	26.328	0.770	0.300
PUP 3D-GS	10%	26.579	**0.785**	**0.274**
QuantileGS (Ours)	10%	**26.849**	0.769	0.303

Gaussian uses volume-adjusted importance pruning and recovery without SH distillation or quantization. All experiments use an NVIDIA RTX 3090 with 24 GB memory.

Results. Table 3 shows that QuantileGS achieves the highest PSNR at both retention levels, although both baselines retain better SSIM and LPIPS. At 20% and 10% retention, the mean Gaussian count decreases from 2.44 million to 0.488 million and 0.244 million. RGB-only PLY storage falls from 577.0 to 115.4 and 57.7 MiB, while RGB rendering throughput increases from 112.7 to 284.4 and 348.4 FPS, respectively. Empirical coverage increases from 92.14% to 96.44% and 96.72% at 20% and 10% retention, respectively, while mean error bounds widen from 0.0929 to 0.1125 and 0.1159. On the evaluated test pixels, no originally covered pixel becomes uncovered under the transported bounds after pruning and refinement, within a numerical tolerance of 10^{-6}. The increased coverage reflects more conservative transported bounds, whose evaluation additionally requires the original model and its quantile fields. These additional costs are excluded from the reported storage and rendering throughput. At 10% retention, the PSNR drop from the unpruned model is 0.321 dB for QuantileGS, compared with 0.591 dB for PUP 3D-GS and 0.842 dB for LightGaussian. The results show a trade-off between PSNR preservation and structural/perceptual similarity. More experiments are provided in Appendix D.1.

Table 4. Replica tracking results averaged over eight scenes, with seed 0. ATE and RPE are in cm. The best and second-best results are highlighted, with the best also in bold.

Method	ATE ↓	RPE ↓	FPS ↑
Gaussian-SLAM	0.329	0.0878	0.423
VarSplat	**0.237**	0.0517	0.256
SplaTAM	0.393	0.0689	0.135
QuantileGS (Ours)	0.290	**0.0046**	**1.058**

5.2.3. Simultaneous Localization and Mapping

Setting and baselines. We evaluate on eight Replica RGB-D sequences (room0-2 and office0-4), using all 2,000 frames at 1200×680 resolution, stride 1, and seed 0. All experiments use an NVIDIA RTX 3090 with 24 GB memory. We reproduce Gaussian-SLAM [48], VarSplat [37], and SplaTAM [24] with method-specific settings. Gaussian-SLAM and VarSplat initialize from two ground-truth poses, while SplaTAM and QuantileGS use no ground-truth motion. QuantileGS uses up to 40 tracking iterations, 120 initialization iterations, and 60 mapping iterations. Mapping runs every five frames with an eight-keyframe window and a one-million-Gaussian limit. Each successful keyframe uses four UQ fitting epochs with projected Adam and a 5% RGB-pixel holdout. We retain the background term and gap normalization, targeting 90% coverage with an online feedback step size of 0.01. We report equal-scene mean ATE and adjacent-frame translational RPE as RMSE, retaining all frames and using SE(3) alignment without scale correction. FPS includes initialization, tracking, mapping, UQ, and online feedback, excluding input decoding and evaluation.

Results. Table 4 shows that QuantileGS achieves the lowest translational RPE and highest throughput, with the second-lowest ATE after VarSplat. Online feedback improves mean pre-update RGB holdout coverage from 87.87% to 89.59% across 399 evaluated keyframes per sequence, approaching the 90% target with nearly unchanged throughput. These results assess camera tracking and empirical RGB coverage, rather than global reconstruction quality or trajectory confidence. Relative to VarSplat, QuantileGS reduces mean translational RPE by approximately 91.1%, while its ATE is 0.053 cm higher. This distinction indicates stronger adjacent-frame translational consistency without the lowest globally aligned trajectory error. More experiments are provided in Appendix D.2.

6. Conclusion

We presented QuantileGS, a post-hoc UQ framework that learns ordered error quantiles for a frozen 3DGS renderer. Scale-normalized conformal calibration yields spatially adaptive RGB-MAE upper bounds with finite-sample marginal coverage under the stated assumptions. Experiments on LF, LLFF, and Mip-NeRF 360 show improved error ranking over the evaluated baselines. Calibration reduces empirical undercoverage on LLFF and Mip-NeRF 360, while LF bounds remain conservative. Applications to active view selection, Gaussian pruning, and RGB-D SLAM empirically demonstrate the downstream utility of the calibrated bounds.

LLM Usage Statement

We used ChatGPT for language polishing and grammatical correction. All technical content and conclusions were developed and verified solely by the authors.

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QuantileGS: Quantile-Based Conformal Bounds on RGB Rendering Errors for Novel View Synthesis with 3D Gaussian Splatting

Supplementary Material

A. Proof of Theorem 1

A.1. Joint Local-Scale Assumptions

Condition on $\mathcal{F}_{\text{fit}}$, comprising all information from $\mathcal{V}_{\text{fit}}$ used to construct the frozen RGB predictor, quantile fields, and β, including independent optimization randomness. The level τ and floor $\varepsilon > 0$ are fixed before using $\mathcal{V}_{\text{cal}}$. Index its n calibration pixels by $i = 1, \dots, n$ and one new test observation by $i = n+1$. For functions μ, σ fixed under this conditioning, assume

$$\begin{aligned} &E_i = \mu(X_i) + \sigma(X_i) Z_i, \sigma(X_i) > 0, \\ &Z_i \overset{\text{i.i.d.}}{\sim} F_Z, i = 1, \dots, n+1. \end{aligned} \tag{A.1}$$

The entire residual vector $(Z_1, \dots, Z_{n+1})$ is independent of the ray vector $(X_1, \dots, X_{n+1})$. The fitted fields share the same location and scale:

$$\hat{q}_\nu(\boldsymbol{x}) = \mu(\boldsymbol{x}) + \sigma(\boldsymbol{x})\hat{z}_\nu, \nu \in \{\eta, \tau\}, \tag{A.2}$$

where $\hat{z}_\eta < \hat{z}_\tau$ are finite constants determined by fitting. Finally, almost surely,

$$\Delta(X_i) = \sigma(X_i)(\hat{z}_\tau - \hat{z}_\eta) \geq \varepsilon, i = 1, \dots, n+1. \tag{A.3}$$

This inactive-floor condition is required for the scale cancellation used below. The three-way view split does not itself establish these distributional assumptions.

A.2. Scale Cancellation and Rank Argument

Proof. All distributions below are conditional on $\mathcal{F}_{\text{fit}}$. After establishing score exchangeability, we apply the standard split-conformal rank argument [26, 33]. For calibration observations $i = 1, \dots, n$ and the test observation $i = n+1$, define the random normalized scores

$$S_i = \frac{E_i - \hat{q}_\tau(X_i)}{D_\varepsilon(X_i)}, i = 1, \dots, n+1. \tag{A.4}$$

Condition (A.3) gives $D_\varepsilon(X_i) = \Delta(X_i)$ almost surely. Substituting Eqs. (A.1) and (A.2) therefore yields

$$\begin{aligned} S_i &= \frac{\sigma(X_i)(Z_i - \hat{z}_\tau)}{\sigma(X_i)(\hat{z}_\tau - \hat{z}_\eta)} \\ &= \frac{Z_i - \hat{z}_\tau}{\hat{z}_\tau - \hat{z}_\eta} =: h(Z_i). \end{aligned} \tag{A.5}$$

The map h is common to all observations and is strictly increasing because $\hat{z}_\tau - \hat{z}_\eta > 0$. It removes all ray-dependent location and scale terms. For arbitrary real thresholds $t_1, \dots, t_{n+1}$, conditional independence of the standardized residuals gives

$$\begin{aligned} &\mathbb{P}\{S_i \leq t_i,\ i = 1, \dots, n+1 \mid \mathcal{F}_{\text{fit}}\} \\ &\quad = \prod_{i=1}^{n+1} F_Z(\hat{z}_\tau + (\hat{z}_\tau - \hat{z}_\eta)t_i)\,. \end{aligned} \tag{A.6}$$

The residual vector is also independent of the ray vector conditional on fitting, so the same joint distribution holds when additionally conditioning on the rays. Thus the scores are conditionally i.i.d. with a common ray-invariant distribution, and hence jointly exchangeable.

First suppose that $k_\tau = \lceil (n+1)\tau \rceil \leq n$. Let $S_{(k_\tau)}$ and $Z_{(k_\tau)}$ denote the k_τ-th order statistics of the n calibration scores and standardized residuals, respectively. Since h is strictly increasing, the calibration rule is

$$\hat{c}_\tau = S_{(k_\tau)} = h(Z_{(k_\tau)}). \tag{A.7}$$

For any ray $\boldsymbol{x}$ on which $\Delta(\boldsymbol{x}) \geq \varepsilon$, substituting this coefficient into Eq. (18) gives

$$\begin{aligned} \hat{u}_\tau(\boldsymbol{x}) &= \mu(\boldsymbol{x}) + \sigma(\boldsymbol{x})\hat{z}_\tau \\ &\quad + \frac{Z_{(k_\tau)} - \hat{z}_\tau}{\hat{z}_\tau - \hat{z}_\eta}\sigma(\boldsymbol{x})(\hat{z}_\tau - \hat{z}_\eta) \\ &= \mu(\boldsymbol{x}) + \sigma(\boldsymbol{x})Z_{(k_\tau)}. \end{aligned} \tag{A.8}$$

In particular, this identity holds at the test ray almost surely by Eq. (A.3).

Let R_{n+1} be the rank of the test score among all $n+1$ scores. Conditional on fitting, break ties using i.i.d. $\text{Uniform}(0,1)$ auxiliary variables that are independent of all observations. Exchangeability and this symmetric tie-breaking rule imply

$$\mathbb{P}\{R_{n+1} = r \mid \mathcal{F}_{\text{fit}}\} = \frac{1}{n+1}, r = 1, \dots, n+1. \tag{A.9}$$

The event $R_{n+1} \leq k_\tau$ implies $S_{n+1} \leq S_{(k_\tau)} = \hat{c}_\tau$. Consequently,

$$\begin{aligned} &\mathbb{P}\{S_{n+1} \leq \hat{c}_\tau \mid \mathcal{F}_{\text{fit}}\} \\ &\quad \geq \mathbb{P}\{R_{n+1} \leq k_\tau \mid \mathcal{F}_{\text{fit}}\} \\ &\quad = \frac{k_\tau}{n+1} \geq \tau. \end{aligned} \tag{A.10}$$

Ties can enlarge the non-strict threshold event, so no continuity assumption on F_Z is needed for this lower bound.

Since $D_\varepsilon(X_{n+1}) > 0$, the score inequality is equivalent to the upper-error event:

$$\begin{aligned}\{S_{n+1} \le \hat{c}_\tau\} &= \{E_{n+1} \le \hat{q}_\tau(X_{n+1}) + \hat{c}_\tau D_\varepsilon(X_{n+1})\} \\ &= \{E_{n+1} \le \hat{u}_\tau(X_{n+1})\}.\end{aligned} \tag{A.11}$$

Combining Eqs. (A.10) and (A.11) proves Eq. (19) for $k_\tau \le n$.

If $k_\tau > n$, Eq. (17) sets $\hat{c}_\tau = +\infty$. Because $D_\varepsilon(X_{n+1}) \ge \varepsilon > 0$, the unclipped upper prediction is $+\infty$ and coverage is one. The clipped RGB-MAE bound is then one, which also covers the error because $E_{n+1} \in [0, 1]$.

This completes the proof. □

A.3. When the Scale Floor Is Active

Under the same shared location-scale model, the actual score is

$$S_i = \frac{Z_i - \hat{z}_\tau}{\max\{\hat{z}_\tau - \hat{z}_\eta, \varepsilon/\sigma(X_i)\}}. \tag{A.12}$$

An active floor can reintroduce ray-dependent scale. The remaining local-scale assumptions therefore need not give exchangeable scores. If exchangeability of the actual floored scores holds separately, the rank argument still proves marginal coverage because $D_\varepsilon > 0$. The guarantee does not assert coverage at every fixed ray or simultaneous coverage of all pixels in a view.

B. Downstream Application Details

This section presents the objectives and computations for the three applications introduced in Section 4.5.

B.1. Active View Selection

B.1.1. View Selection Criterion

AVS selects camera viewpoints to improve scene reconstruction under a limited acquisition budget. Building on the local information approximation of FisherRF [23], we prioritize views that supply missing information about parameters affecting regions with large calibrated error bounds.

At each round, we freeze 3DGS, fit QuantileGS on acquired and reserved fitting views, and calibrate on a disjoint subset. The bounds $u_v(p) \in [0, 1]$ include the background contribution and gap normalization. Reserved UQ views are excluded from reconstruction training and the candidate pool.

Let $\mathcal{V}_0, \mathcal{V}_c$ denote reconstruction and candidate views, and let $\boldsymbol{\theta} \in \mathbb{R}^d$ collect Gaussian geometry, opacity, and active SH parameters. For view v with N_v pixels, set $n_v = 3N_v$ and stack rendered RGB into $\boldsymbol{C}_v \in \mathbb{R}^{n_v}$. Let $J_v = \partial \boldsymbol{C}_v / \partial \boldsymbol{\theta} \in \mathbb{R}^{n_v \times d}$ be the RGB Jacobian at the current model, before output truncation to $[0, 1]$. The diagonal matrix $W_v \in \mathbb{R}^{n_v \times n_v}$ repeats $u_v(p)$ across RGB channels. It assigns prediction priorities rather than noise variances. Define

$$\begin{aligned}\boldsymbol{F}_v &= \frac{\operatorname{diag}(J_v^\top J_v)}{n_v}, \\ \boldsymbol{\Lambda} &= \sum_{s \in \mathcal{V}_0} \boldsymbol{F}_s, \\ \boldsymbol{\omega} &= \frac{1}{|\mathcal{V}_c|} \sum_{r \in \mathcal{V}_c} \frac{\operatorname{diag}(J_r^\top W_r J_r)}{n_r}.\end{aligned} \tag{B.1}$$

For parameter $k = 1, \dots, d$, $F_{v,k}$ measures candidate information, Λ_k summarizes acquired information, and ω_k measures influence on high-bound regions.

Using a normalized least-squares surrogate with curvature $J_v^\top J_v / n_v$, a diagonal local Gaussian approximation gives the parameter-covariance proxy $\Sigma(\boldsymbol{\Lambda}) = \operatorname{diag}((\Lambda_k + \lambda)^{-1})$, with regularization $\lambda > 0$. A first-order rendering approximation propagates it to $J_r \Sigma(\boldsymbol{\Lambda}) J_r^\top$. Its error-weighted mean trace is

$$\begin{aligned}\mathcal{P}_u(\boldsymbol{\Lambda}) &= \frac{1}{|\mathcal{V}_c|} \sum_{r \in \mathcal{V}_c} \frac{\operatorname{tr}(W_r J_r \Sigma(\boldsymbol{\Lambda}) J_r^\top)}{n_r} \\ &= \sum_{k=1}^{d} \frac{\omega_k}{\Lambda_k + \lambda}.\end{aligned} \tag{B.2}$$

We replace FisherRF's parameter-information objective with the reduction of this error-weighted prediction-variance proxy. Within a frozen round, candidate v contributes $\boldsymbol{F}_v$ and receives the score

$$\begin{aligned}\mathcal{A}(v) &= \mathcal{P}_u(\boldsymbol{\Lambda}) - \mathcal{P}_u(\boldsymbol{\Lambda} + \boldsymbol{F}_v) \\ &= \sum_{k=1}^{d} \frac{\omega_k F_{v,k}}{(\Lambda_k + \lambda)(\Lambda_k + \lambda + F_{v,k})}.\end{aligned} \tag{B.3}$$

We acquire the highest-scoring view and recompute all quantities after RGB retraining. Reconstruction gains are evaluated empirically and do not follow from the rendering-error coverage guarantee.

B.1.2. Implementation Details

We estimate the unweighted and weighted diagonals using 32 Rademacher vector–Jacobian probes per view, applying $W_v^{1/2}$ to the weighted probes and holding the bounds fixed during differentiation. We set λ to 0.01 times the mean positive entry of $\boldsymbol{\Lambda}$. Scoring requires no candidate RGB observations.

After each acquisition, we retrain 3DGS from the same initialization, progressing from four initial views to 20 reconstruction views. Each round refits QuantileGS using projected Adam for 200 epochs at learning rate 0.01. RGB training uses random backgrounds, while UQ fitting, calibration, scoring, and evaluation use fixed black backgrounds. Reserved UQ views are additional observations.

B.2. Gaussian Pruning

B.2.1. Construction of Pruning Score

Gaussian pruning aims to reduce the number of Gaussians while maintaining rendering quality. Let $\|\cdot\|$ be the calibrated error norm and $\|\cdot\|_{\mathrm{P}}$ the norm whose square defines the pruning loss. Fix $\kappa > 0$ such that $\|\boldsymbol{z}\|_{\mathrm{P}} \le \kappa\|\boldsymbol{z}\|$. This separates the calibrated error scale from the pruning criterion through a single conversion factor.

Pruning and refinement change the RGB predictor. Let $\hat{\boldsymbol{y}}_0$ and $\hat{\boldsymbol{y}}_t$ be its original and current predictions at the same ray, and let $u^{(0)}$ be the original calibrated bound. We update it as

$$u^{(t)} = \min\left\{e_{\max},\, u^{(0)} + \|\hat{\boldsymbol{y}}_t - \hat{\boldsymbol{y}}_0\|\right\}. \tag{B.4}$$

The triangle inequality and the error limit $e_{\max}$ give

$$\|\boldsymbol{y} - \hat{\boldsymbol{y}}_0\| \le u^{(0)} \implies \|\boldsymbol{y} - \hat{\boldsymbol{y}}_t\| \le u^{(t)}. \tag{B.5}$$

Thus, the update preserves originally covered events at the same observation, including after data-dependent refinement.

For Gaussian j, define its deletion perturbation $\boldsymbol{\delta}_j = \hat{\boldsymbol{y}}_t^{(-j)} - \hat{\boldsymbol{y}}_t$ and $d_j = \|\boldsymbol{\delta}_j\|_{\mathrm{P}}$. Denoting $\boldsymbol{r}_t = \hat{\boldsymbol{y}}_t - \boldsymbol{y}$, on the covered event $\|\boldsymbol{r}_t\| \le u^{(t)}$, the triangle inequality bounds the loss increment by

$$\begin{aligned}
&\|\boldsymbol{r}_t + \boldsymbol{\delta}_j\|_{\mathrm{P}}^2 - \|\boldsymbol{r}_t\|_{\mathrm{P}}^2 \\
&\quad\le d_j^2 + 2\|\boldsymbol{r}_t\|_{\mathrm{P}} d_j \\
&\quad\le d_j^2 + 2\kappa u^{(t)} d_j.
\end{aligned} \tag{B.6}$$

For a fixed deletion perturbation d_j, a larger error bound $u^{(t)}$ increases the estimated deletion cost.

To define the pruning score, we average the pointwise deletion-cost envelope over reference fitting views $\mathcal{V}_{\mathrm{ref}}$. For any pixel quantity $a(v,p)$, define

$$\mathbb{E}_{\mathrm{ref}}[a] = \frac{1}{|\mathcal{V}_{\mathrm{ref}}|}\sum_{v\in\mathcal{V}_{\mathrm{ref}}}\frac{1}{N_v}\sum_{p=1}^{N_v} a(v,p), \tag{B.7}$$

where N_v is the pixel count of view v. The average envelope contains a squared perturbation term and a residual-dependent term. We summarize them using

$$Q_j = \mathbb{E}_{\mathrm{ref}}[d_j^2],\; H_j = \mathbb{E}_{\mathrm{ref}}[(u^{(t)})^2 d_j^2]. \tag{B.8}$$

where Q_j measures the average squared rendering change, while H_j increases when this change affects pixels with large error bounds. Since $\mathbb{E}_{\mathrm{ref}}[1] = 1$, Cauchy-Schwarz applied to $(u^{(t)}d_j)\cdot 1$ yields $\mathbb{E}_{\mathrm{ref}}[u^{(t)}d_j] \le \sqrt{H_j}$.

Consequently,

$$\begin{aligned}
&\mathbb{E}_{\mathrm{ref}}[d_j^2 + 2\kappa u^{(t)} d_j] \\
&\quad= Q_j + 2\kappa\mathbb{E}_{\mathrm{ref}}[u^{(t)} d_j] \\
&\quad\le Q_j + 2\kappa\sqrt{H_j} =: S_j^{\mathrm{prune}}.
\end{aligned} \tag{B.9}$$

We therefore use S_j^{prune} to rank deletions, prioritizing Gaussians with the smallest scores.

This favors small rendering changes while accounting for the unknown residual through the calibrated error scale. In particular, for RGB-MAE calibration and RGB-MSE pruning, $\|\boldsymbol{z}\| = \|\boldsymbol{z}\|_1/3$ and $\|\boldsymbol{z}\|_{\mathrm{P}} = \|\boldsymbol{z}\|_2/\sqrt{3}$, so $\kappa = \sqrt{3}$ and $S_j^{\mathrm{prune}} = Q_j + 2\sqrt{3H_j}$.

B.2.2. Implementation Details

Computing Q_j and H_j requires evaluating the deletion perturbation for every Gaussian, which is expensive. We next estimate these scores without rendering a separate scene for every possible deletion. We temporarily multiply each Gaussian's opacity by a gate g_j: zero removes the Gaussian, while one keeps its original opacity. We evaluate all gates at a common value $\gamma \in (0,1)$. Decreasing gate j from γ to zero gives the first-order deletion approximation

$$\boldsymbol{\delta}_j \approx -\gamma\frac{\partial\hat{\boldsymbol{y}}}{\partial g_j}. \tag{B.10}$$

The gate derivative therefore supplies a local estimate of how much removing the Gaussian changes the image.

For a reference view with N pixels, we render the RGB image once and independently assign $z_{p,c} = -1$ or $+1$ with equal probability to every pixel and color channel. Then we form two scalar sums:

$$\begin{aligned}
L_{\boldsymbol{z}} &= \frac{1}{\sqrt{3N}}\sum_{p,c} z_{p,c}\hat{y}_{p,c}, \\
L_{\boldsymbol{z},u} &= \frac{1}{\sqrt{3N}}\sum_{p,c} u^{(t)}(p) z_{p,c}\hat{y}_{p,c}.
\end{aligned} \tag{B.11}$$

The first randomly adds or subtracts RGB values; the second also weights each pixel by its error bound. Keep these bounds fixed and backpropagate each scalar through the renderer. Each backward pass returns a derivative for every gate, allowing all Gaussians to be scored together.

To derive the moment estimates, we fix the rendered scene and pixel bounds, and write the RGB derivatives at $g_j = \gamma$ as $a_{p,c,j} = \partial\hat{y}_{p,c}/\partial g_j$. Differentiating Eq. (B.11) gives

$$\begin{aligned}
\frac{\partial L_{\boldsymbol{z}}}{\partial g_j} &= \frac{1}{\sqrt{3N}}\sum_{p,c} z_{p,c}a_{p,c,j}, \\
\frac{\partial L_{\boldsymbol{z},u}}{\partial g_j} &= \frac{1}{\sqrt{3N}}\sum_{p,c} u^{(t)}(p) z_{p,c}a_{p,c,j}.
\end{aligned} \tag{B.12}$$

Only the signs are random in this calculation. Each sign has mean zero and variance one, so independence gives $\mathbb{E}_{\boldsymbol{z}}[z_{p,c}z_{p',c'}] = 0$ for distinct pixel-channel pairs. Expand-

ing the first squared gradient therefore yields

$$\begin{aligned}\mathbb{E}_{\boldsymbol{z}}\left[\left(\frac{\partial L_{\boldsymbol{z}}}{\partial g_j}\right)^2\right] \\ = \frac{1}{3N}\sum_{p,c}\sum_{p',c'} a_{p,c,j}a_{p',c',j}\mathbb{E}_{\boldsymbol{z}}[z_{p,c}z_{p',c'}] \\ = \frac{1}{3N}\sum_{p,c} a_{p,c,j}^2.\end{aligned} \tag{B.13}$$

All cross-terms vanish, leaving the sum of squared RGB derivatives. The same argument with the fixed pixel weights gives

$$\begin{aligned}\mathbb{E}_{\boldsymbol{z}}\left[\left(\frac{\partial L_{\boldsymbol{z},u}}{\partial g_j}\right)^2\right] \\ = \frac{1}{3N}\sum_{p,c}(u^{(t)}(p))^2 a_{p,c,j}^2.\end{aligned} \tag{B.14}$$

By Eq. (B.10), multiplying these expectations by γ^2 gives the unweighted and bound-weighted moments of the linearized deletion response. The factor $1/(3N)$ already averages over RGB channels and pixels. We approximate each expectation by averaging K independent random draws, giving the per-view estimates

$$\begin{aligned}\widehat{Q}_{j,v} &= \frac{\gamma^2}{K}\sum_{k=1}^{K}\left(\frac{\partial L_{\boldsymbol{z}_k}}{\partial g_j}\right)^2, \\ \widehat{H}_{j,v} &= \frac{\gamma^2}{K}\sum_{k=1}^{K}\left(\frac{\partial L_{\boldsymbol{z}_k,u}}{\partial g_j}\right)^2.\end{aligned} \tag{B.15}$$

Finally, average these estimates equally over reference views and substitute them into Eq. (B.9) to rank Gaussians. Algorithm 1 summarizes the computation.

We alternate deletion of low-scoring Gaussians with refinement and rescoring as visibility changes. The scores use approximate deletion responses, whereas the bound update in Eq. (B.4) uses the actual RGB predictions before and after pruning.

Algorithm 1 Gaussian Pruning Score Estimation

Input: Frozen Gaussian scene $\mathcal{G}$; reference views $\mathcal{V}_{\text{ref}}$; fixed RGB-MAE bounds $\{u_v^{(t)}\}$; gate value $\gamma \in (0,1)$; number of probes $K \geq 1$.
Output: Pruning scores $\{S_j^{\text{prune}}\}$.

```
Set g_j ← γ for every Gaussian j.
Initialize Q̂_j ← 0, Ĥ_j ← 0 for all j.
for each view v ∈ V_ref do
    Render its RGB image once using the opacity gates.
    for k = 1, ..., K do
        Draw independent signs z_{p,c} ∈ {−1, +1} uniformly.
        Form L_z and L_{z,u} using Eq. (B.11).
        Backpropagate each scalar to all gates.
        For all j, accumulate Q̂_j ← Q̂_j + γ²(∂L_z/∂g_j)².
        For all j, accumulate Ĥ_j ← Ĥ_j + γ²(∂L_{z,u}/∂g_j)².
    end for
end for
for each Gaussian j do
    Q̂_j ← Q̂_j/(K|V_ref|).
    Ĥ_j ← Ĥ_j/(K|V_ref|).
    S_j^prune ← Q̂_j + 2√(3Ĥ_j).
end for
return {S_j^prune}; smaller scores are pruned first.
```

B.3. Simultaneous Localization and Mapping

B.3.1. RGB-D Tracking and Mapping

SLAM estimates a camera trajectory while reconstructing the scene observed along that trajectory. Given RGB images $\boldsymbol{y}_t$, depth maps D_t, and known camera intrinsics $\boldsymbol{K}$, which specify camera projection, we estimate camera poses $T_t \in \mathrm{SE}(3)$ and incrementally build a Gaussian map $\mathcal{G}$. Each pose transforms 3D points from camera to world coordinates. Tracking aligns each incoming frame with the previously observed scene to estimate its pose; mapping then uses this pose to update scene geometry and appearance.

We adopt the alternating tracking-mapping organization of Gaussian-SLAM [48]. Gaussian-SLAM estimates poses against a rendered Gaussian submap, while VarSplat [37] additionally uses learned appearance variance to weight photometric residuals. Our integration uses QuantileGS to construct reference images with calibrated rendering-error bounds. New frames are aligned with these cached RGB-D references by reprojection, and their bounds determine the robust scales of the photometric tracking loss. Thus, Gaussian rendering supplies the reference appearance, while pose optimization operates on image correspondences.

We reserve an RGB pixel subset in each frame for calibration; only the remaining *fitting pixels* enter RGB tracking, map fitting, quantile learning, and reference correction. Depth remains available for geometric alignment. We set $T_0 = I$ and initialize $\mathcal{G}$ from the first frame's fitting RGB-D. Selected frames (*keyframes*) supply reference observations for subsequent tracking and map refinement. Let $\mathcal{W}$ denote a recent-keyframe window used for fitting, and let $\mathcal{R}$ index the recent reference keyframes used for tracking. The input frame-index sets $\mathcal{K}$ and $\mathcal{M}$ specify keyframes and map-refinement frames, respectively, with $0 \in \mathcal{K}$.

B.3.2. Calibrated Reference Construction

After updating the map at keyframe k, we freeze a copy, fit the ordered quantile fields on fitting RGB errors from $\mathcal{W}$, and calibrate them on the fresh reserved RGB of frame k. Rendering this frozen predictor at T_k gives colors $\hat{\boldsymbol{y}}_{k,p}$ and bounds $u_{k,p}$ on $\|\boldsymbol{y}_{k,p} - \hat{\boldsymbol{y}}_{k,p}\|$. We then incorporate the observed fitting colors into the saved reference, using the bound as a correction budget. This allows the reference

to use measured appearance while controlling its departure from the calibrated rendering.

Similar with Section B.2, we choose κ_{ch} such that $|a_c| \le \kappa_{\mathrm{ch}}\|\boldsymbol{a}\|$ for every RGB error vector $\boldsymbol{a}$ and channel c. A covered RGB error therefore implies a channel error at most $b_{k,p} = \kappa_{\mathrm{ch}} u_{k,p}$. At fitting pixels, the corrected reference is

$$\widetilde{y}_{k,p,c} = \hat{y}_{k,p,c} + \Pi_{[-b_{k,p},\, b_{k,p}]}(y_{k,p,c} - \hat{y}_{k,p,c}), \tag{B.16}$$

where $\Pi_I(v) = \arg\min_{s\in I}(s-v)^2$ denotes projection onto interval I. Intuitively, each channel moves toward its observation by at most $b_{k,p}$; on the covered event it reaches that observation. Reserved pixels retain their rendered colors, and the Gaussian parameters remain fixed. As in Eq. (B.4), we attach the conservatively transported bound

$$\widetilde{u}_{k,p} = \min\left\{e_{\max},\, u_{k,p} + \|\widetilde{\boldsymbol{y}}_{k,p} - \hat{\boldsymbol{y}}_{k,p}\|\right\}. \tag{B.17}$$

The triangle inequality preserves every originally covered observation under this change of reference. We cache $(\widetilde{\boldsymbol{y}}_k, \widetilde{u}_k, D_k, T_k)$ and the fitting mask; subsequent map updates leave this reference fixed. Our implementation uses RGB-MAE, for which $\kappa_{\mathrm{ch}} = 3$ and $e_{\max} = 1$.

B.3.3. Uncertainty-Aware Tracking and Map Updates

For an incoming frame t, we estimate its pose by aligning its observed RGB-D with the references in $\mathcal{R}$. Depth converts pixel p into a 3D point, and a candidate pose T projects that point into reference image k:

$$\begin{aligned} \boldsymbol{X}_t(p) &= D_t(p)\boldsymbol{K}^{-1}\bar{\boldsymbol{p}}, \\ w_{k\leftarrow t}(p;T) &= \pi_{\boldsymbol{K}}\big(T_k^{-1}T\boldsymbol{X}_t(p)\big), \end{aligned} \tag{B.18}$$

where p_x, p_y are the horizontal and vertical pixel coordinates, $\bar{\boldsymbol{p}} = (p_x, p_y, 1)^\top$, and $\pi_{\boldsymbol{K}}$ denotes camera projection. Where $\boldsymbol{X}_t(p)$ is expressed in the current camera's coordinates; $T_k^{-1}T$ transforms it into reference-camera coordinates. The resulting pixel location $w_{k\leftarrow t}(p;T)$ determines which reference color to compare with the current observation:

$$r_{k,p,c}(T) = \widetilde{y}_{k,c}\big(w_{k\leftarrow t}(p;T)\big) - y_{t,p,c}. \tag{B.19}$$

The error bound at the matched reference pixel determines how strongly this color residual influences tracking. We set its robust scale $\ell_{k,p}$ from $\kappa_{\mathrm{ch}}\widetilde{u}_k(w_{k\leftarrow t}(p;T))$, with a positive floor to avoid zero scales. The photometric objective is

$$\mathcal{L}_{\mathrm{rgb}}(T) = \sum_{k\in\mathcal{R}}\sum_{p\in\Omega_k} a_{k,p}\sum_{c=1}^{3}\rho_{\mathrm{H}}\left(\frac{r_{k,p,c}(T)}{\ell_{k,p}}\right), \tag{B.20}$$

where Ω_k contains visible, depth-consistent fitting matches, and $a_{k,p} \ge 0$ combines fitting-pixel weights with per-reference normalization.

The Huber penalty $\rho_{\mathrm{H}}(z)$ equals $z^2/2$ for $|z| \le 1$ and $|z| - 1/2$ otherwise. Thus, the calibrated bound supplies a local error scale, while the linear tail limits the influence of large photometric mismatches. Reserved RGB is excluded before image resizing or sampling.

Depth provides a complementary geometric constraint. We use $\mathcal{L}_{\mathrm{depth}}(T)$, a robust point-to-plane alignment cost: it penalizes the distance between transformed current depth points and the corresponding reference surfaces. This term uses observed depth and a separate scale in meters; RGB error bounds are used only in the photometric term. We estimate T_t by minimizing

$$\mathcal{L}_{\mathrm{track}}(T) = \mathcal{L}_{\mathrm{rgb}}(T) + \lambda_d\mathcal{L}_{\mathrm{depth}}(T), \lambda_d > 0. \tag{B.21}$$

We initialize from recent camera motion and apply Gauss-Newton updates from coarse to fine. We refresh matches and sampled scales between iterations, keeping masks, scales, and weights fixed when testing an update.

After tracking, accepted frames in $\mathcal{M}\cup\mathcal{K}$ update $\mathcal{G}$ with the estimated poses fixed. Map refinement minimizes fitting color and depth errors over the current frame and $\mathcal{W}$. At accepted keyframes, newly observed fitting RGB-D initializes additional Gaussians before refinement, and a calibrated reference is constructed afterward for future tracking, as shown in Algorithm 2. The calibration statement concerns rendering errors, and we evaluate its effect on trajectory accuracy empirically.

In Algorithm 2, $\mathcal{F}_t$ and $\mathcal{H}_t$ denote fitting and reserved RGB pixels. MapUpdate performs the map update described above, and Recent maintains the corresponding recent window. For a score set S of size n, $\mathsf{Q}^+_{1-a}(S)$ denotes the $\lceil(n+1)(1-a)\rceil$-th order statistic of $S\cup\{+\infty\}$. Within reference construction, pixel-wise quantities refer to the current frame.

B.3.4. Online Calibration

We adapt the calibration level using observed coverage as the camera moves, and the map evolves. The nominal miscoverage level remains fixed at $\alpha = 0.1$. We maintain an effective calibration level a_t, initialized with $a_0 = \alpha$.

After tracking an evaluated keyframe t, but before updating the map or quantile fields, we measure coverage on its reserved RGB pixels $\mathcal{H}_t$:

$$\widehat{C}_t = \frac{1}{|\mathcal{H}_t|}\sum_{p\in\mathcal{H}_t}\mathbf{1}\left\{\|\hat{\boldsymbol{y}}^-_{t,p} - \boldsymbol{y}_{t,p}\| \le u^-_{t,p}\right\}. \tag{B.22}$$

where $\hat{\boldsymbol{y}}^-_{t,p}$ is rendered from the map before its update, and $u^-_{t,p}$ is supplied by the existing calibrated predictor, transported to this map using Eq. (B.4) when necessary.

We update the effective miscoverage level by

$$a_{t+1} = \Pi_{[a_{\min},a_{\max}]}\Big(a_t + \gamma\big[\widehat{C}_t - (1-\alpha)\big]\Big), \tag{B.23}$$

Algorithm 2 QuantileGS RGB-D SLAM with Online Calibration

Input: RGB-D frames $\{(\boldsymbol{y}_t, D_t, \mathcal{F}_t, \mathcal{H}_t)\}$, intrinsics $\boldsymbol{K}$; quantile levels (η, τ) and floor ϵ; nominal miscoverage α; feedback parameters γ, $[a_{\min}, a_{\max}]$; frame sets $\mathcal{K}, \mathcal{M}$.
Output: Camera poses $\{T_t\}$ and Gaussian map $\mathcal{G}$.

$\mathcal{G}, \mathcal{W}, \mathcal{R} \leftarrow \varnothing$; $T_0 \leftarrow I$; $a \leftarrow \alpha$.
for each frame t **do**
  For $t > 0$, estimate T_t against $\mathcal{R}$ by minimizing $\mathcal{L}_{\text{rgb}}(T) + \lambda_d \mathcal{L}_{\text{depth}}(T)$.
  If tracking is rejected, $T_t \leftarrow T_{t-1}$.
  if $t \in \mathcal{K}$ and a predictor is available **then**
    $\widehat{C}_t \leftarrow \dfrac{1}{|\mathcal{H}_t|} \sum_{p \in \mathcal{H}_t} \mathbf{1}\{\|\hat{\boldsymbol{y}}^-_{t,p} - \boldsymbol{y}_{t,p}\| \le u^-_{t,p}\}$.
    $a \leftarrow \Pi_{[a_{\min}, a_{\max}]}\big(a + \gamma[\widehat{C}_t - (1-\alpha)]\big)$.
  end if
  if $t = 0$ or tracking is accepted **then**
    If $t \in \mathcal{M} \cup \mathcal{K}$, $\mathcal{G} \leftarrow \mathsf{MapUpdate}(\mathcal{G}, \mathcal{W}, t)$.
    if $t \in \mathcal{K}$ **then**
      $\mathcal{W} \leftarrow \mathsf{Recent}(\mathcal{W}, t)$; $\mathcal{G}^\star \leftarrow \mathsf{FreezeCopy}(\mathcal{G})$.
      $\phi \leftarrow \mathsf{FitQuantiles}(\mathcal{G}^\star, \mathcal{W}; \eta, \tau)$ using fitting RGB only.
      Render $(\hat{\boldsymbol{y}}, \hat{q}_\eta, \hat{q}_\tau)$ from $(\mathcal{G}^\star, \phi)$ at T_t.
      $d_p \leftarrow \max\{\hat{q}_{\tau,p} - \hat{q}_{\eta,p}, \epsilon\}$.
      $s_p \leftarrow \dfrac{\|\boldsymbol{y}_{t,p} - \hat{\boldsymbol{y}}_p\| - \hat{q}_{\tau,p}}{d_p}, \quad p \in \mathcal{H}_t$.
      $\hat{c} \leftarrow \mathsf{Q}^+_{1-a}(\{s_p : p \in \mathcal{H}_t\})$.
      $u_p \leftarrow \text{clip}_{[0, e_{\max}]}(\hat{q}_{\tau,p} + \hat{c}\, d_p)$.
      $\widetilde{\boldsymbol{y}} \leftarrow \hat{\boldsymbol{y}}$; $b_p \leftarrow \kappa_{\text{ch}} u_p$.
      For $p \in \mathcal{F}_t$ and each channel c:
      $\widetilde{y}_{p,c} \leftarrow \hat{y}_{p,c} + \Pi_{[-b_p, b_p]}(y_{t,p,c} - \hat{y}_{p,c})$.
      $\widetilde{u}_p \leftarrow \min\{e_{\max},\, u_p + \|\widetilde{\boldsymbol{y}}_p - \hat{\boldsymbol{y}}_p\|\}$.
      $\mathcal{R} \leftarrow \mathsf{Recent}(\mathcal{R}, (\widetilde{\boldsymbol{y}}, \widetilde{u}, D_t, T_t, \mathcal{F}_t))$.
    end if
  end if
end for
return $\{T_t\}, \mathcal{G}$.

where $\gamma = 0.01$, $a_{\min} = 0.001$, and $a_{\max} = 0.5$. On frames without feedback, $a_{t+1} = a_t$. Undercoverage decreases a_t, requesting a higher calibration quantile; overcoverage increases it.

At the next successful quantile-field refresh, the latest effective level replaces α in the conformal order statistic. The fitting levels η, τ remain fixed. The current keyframe's reserved RGB pixels subsequently calibrate the refreshed predictor, which supplies references for future tracking. Feedback does not revise the completed tracking estimate or its recorded pre-update coverage. This mechanism provides empirical coverage control and does not establish a per-frame finite-sample coverage guarantee.

C. Uncertainty Quantification Results

We provide additional results on five selected scenes not shown in the main qualitative comparison: LLFF flower and room, Mip-NeRF 360 bonsai and stump, and LF torch. QuantileGS-U and QuantileGS-C denote predictions before and after conformal calibration, respectively.

C.1. Selected-Scene Quantitative Results

Table 5 reports AUSE and Pearson correlation, retaining each baseline's original evaluation conventions. QuantileGS-C outperforms all baselines in both ranking metrics on the selected LLFF and Mip-NeRF 360 scenes. On torch, it achieves the lowest AUSE alongside QuantileGS-U, while Variational-3DGS retains higher Pearson correlation. Fig. 4 compares RGB renderings, rendering errors, and uncertainty maps on the same five scenes.

C.2. Empirical Coverage and Bound Tightness

We compare QuantileGS-U and QuantileGS-C on all scenes using the same configurations and test views as in the UQ evaluation. Coverage is the fraction of test pixels whose RGB-MAE does not exceed the predicted upper bound. Mean bound averages these predictions over the same pixels. The nominal coverage level is 90%, with $\alpha = 0.1$. Both quantities are computed by pooling test pixels within each scene. Dataset averages weight scenes equally. A smaller mean bound indicates tighter predictions but must be interpreted together with coverage.

Coverage and bound size. Table 6 shows that calibration substantially reduces undercoverage on LLFF and Mip-NeRF 360. Mean coverage increases from 69.56% to 89.61% on LLFF and from 80.07% to 92.15% on Mip-NeRF 360. These gains accompany wider bounds, whose means increase from 0.0413 to 0.0965 and from 0.0513 to 0.0929, respectively. The correction varies across scenes. For example, stump coverage rises from 64.19% to 91.48% as its mean bound increases from 0.0444 to 0.1351. Bonsai requires a smaller adjustment, from 0.0350 to 0.0431, to raise coverage from 87.15% to 91.40%.

Scene-level variation. All nine Mip-NeRF 360 scenes exceed the nominal coverage level after calibration, with empirical coverage ranging from 90.44% to 94.91%. LLFF remains less uniform. Although its mean coverage approaches 90%, six scenes remain below this level, with the largest shortfall on trex at 84.59%. The dataset mean therefore does not establish adequate coverage for every scene.

Conservative predictions on LF. LF already exceeds the nominal level before calibration. Mean coverage changes only slightly, from 96.32% to 96.39%, while the mean

Table 5. UQ results on five additional selected scenes. The best , second-best , and third-best AUSE and Pearson results are highlighted, with the best also in bold.

Method	LLFF				Mip-NeRF 360				LF	
	Flower		Room		Bonsai		Stump		Torch	
	AUSE ↓	Pears. ↑	AUSE ↓	Pears. ↑	AUSE ↓	Pears. ↑	AUSE ↓	Pears. ↑	AUSE ↓	Pears. ↑
Variational-3DGS	0.379	0.226	0.357	0.279	0.473	0.072	0.526	0.123	0.223	**0.434**
FisherRF	0.328	-0.094	0.382	-0.070	0.503	-0.071	0.356	-0.020	0.700	-0.051
PRIMU	0.257	0.406	0.375	0.207	0.364	0.264	0.435	0.134	0.345	0.358
Manifold	0.462	0.182	0.468	0.179	0.350	0.302	0.483	0.144	0.514	0.126
QuantileGS-U (Ours)	**0.144**	**0.458**	0.268	**0.336**	**0.299**	0.340	**0.306**	**0.481**	**0.218**	0.385
QuantileGS-C (Ours)	0.156	0.452	**0.225**	0.333	0.300	**0.343**	0.309	0.450	**0.218**	0.386

bound changes from 0.0684 to 0.0691. Calibration tightens the bounds on basket and statue, whose coverage remains above 90%, but widens them on africa and torch. Thus, calibration does not uniformly enlarge the bounds or bring every scene closer to the nominal level. These results report empirical coverage under the evaluated protocols rather than conditional coverage for individual views or pixels.

D. Downstream Application Results

We provide per-scene results and qualitative comparisons for Gaussian pruning and SLAM. All experiments use seed 0, and dataset averages weight scenes equally. Performance values are ranked before rounding. The best , second-best , and third-best results are highlighted, with the best also in bold. Coverage, bound widths, and model sizes are reported as diagnostic quantities without ranking.

D.1. Gaussian Pruning

Per-scene comparison. We evaluate all nine Mip-NeRF 360 scenes at 20% and 10% retention. All methods use the original checkpoints and test views, and rendering metrics are computed from floating-point RGB images. QuantileGS attains the highest mean PSNR at both retention levels: it ranks first on five scenes at 20% retention and six at 10%, and its mean PSNR exceeds that of PUP by 0.053 and 0.269 dB, respectively.

Model size and transported bounds. Table 7 reports Gaussian counts, QuantileGS coverage, and mean RGB-MAE bounds. Counts are identical across methods at each retention level. The mean count decreases from approximately 2.44 million to 488 thousand and 244 thousand. Coverage increases from 92.14% to 96.44% and 96.72%, while mean bounds widen from 0.0929 to 0.1125 and 0.1159. No originally covered test pixel becomes uncovered under the transported bound in the saved evaluations. The higher coverage accompanies more conservative bounds and does not establish improved calibration or a PSNR guarantee. Evaluating transported bounds additionally requires the original renderer and its quantile fields.

Qualitative comparison. Fig. 5 compares RGB renderings and RGB-MAE maps on the same test views of the kitchen and bicycle scenes at both retention levels. QuantileGS achieves the highest scene-level PSNR at both levels, with its largest gain over PUP occurring at 10% retention.

Calibration and Target Coverage We study the effect of calibration on all nine Mip-NeRF 360 scenes using the original RGB checkpoints and quantile fields from our pruning protocol, evaluated before any pruning or recovery. The RGB renderer and the learned quantile fields remain fixed throughout this analysis. In particular, the fitted quantile levels are $\eta = 0.5$ and $\tau = 0.9$ for every variant, and changing the calibration target does not involve refitting these fields. We reuse the saved fitting, calibration, and test partitions, with 21-54 calibration views per scene, and use seed 0. No test observations are used to determine the calibration corrections.

We compare the uncalibrated upper prediction, denoted by U, with gap-normalized conformal calibration at nominal coverage levels $1 - \alpha \in \{0.80, 0.90, 0.95\}$, denoted by C80, C90, and C95. Coverage and mean bounds are computed by pooling test pixels within each scene and then averaging scenes equally. RGB values are normalized to $[0, 1]$, and bounds are expressed in RGB-MAE units. Table 8 shows that the uncalibrated prediction covers 80.01% of test pixels on average, below the nominal 90% level considered by C90. Calibration raises empirical coverage to 92.14%, accompanied by an increase in the mean bound from 0.0513 to 0.0929. This comparison demonstrates reduced empirical undercoverage relative to the prescribed target.

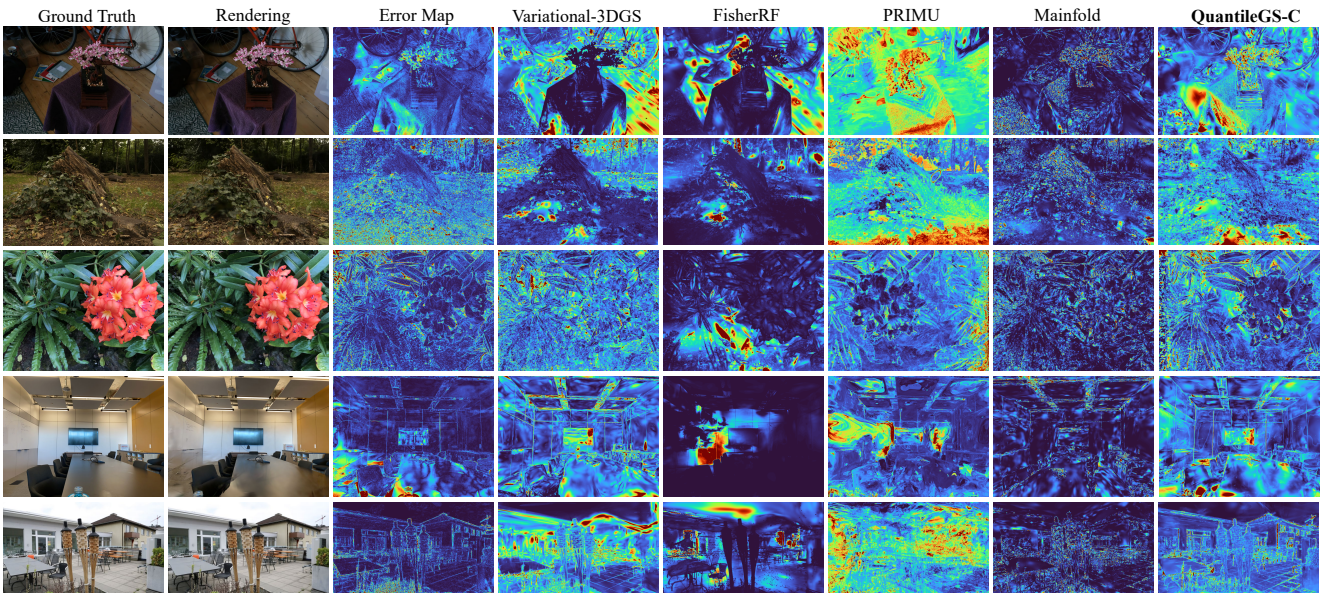

Figure 4. Additional qualitative comparisons of Variational-3DGS [28], FisherRF [23], PRIMU [17], Manifold [30], and QuantileGS-C (Ours). Each scene shows uncertainty maps. Scenes from top to bottom: Mip-NeRF 360 bonsai and stump, LLFF flower and room, and LF torch.

Increasing the nominal target from 80% to 90% and 95% raises empirical coverage from 82.71% to 92.14% and 96.65%, respectively, while the mean bound increases from 0.0593 to 0.0929 and 0.1379. These results illustrate the empirical trade-off between coverage and bound tightness: higher coverage requirements produce more conservative bounds, without changing the fitted quantile fields. Since calibration leaves the RGB renderer unchanged, all variants produce identical RGB images and rendering-quality metrics for the original model. This property does not imply unchanged performance when the bounds guide subsequent downstream decisions.

D.2. Simultaneous Localization and Mapping

Role of coverage in tracking. Rendering-error bounds determine both reference correction budgets and photometric residual scales. For RGB-MAE, a covered error implies a per-channel error of at most $3u_{k,p}$. Thus, at a covered fitting pixel, the budget in Eq. (B.16) allows full correction to the observed color. The transported bound subsequently sets the robust scale of the photometric tracking loss. Coverage therefore provides an explicit criterion for assessing the rendering-error bounds used by these decisions.

Uncertainty and calibration ablation. We compare No-UQ, which disables reference correction and uses unweighted L1 photometric tracking, with U, which uses uncalibrated bounds for reference correction and photometric residual scales. Online reuses the online-calibrated results reported above. Table 9 and Fig. 6 summarize the comparison. Relative to No-UQ, U reduces mean ATE and RPE by 18.3% and 56.6%, improving six and eight scenes, respectively. This supports the joint use of uncertainty in reference construction and tracking. Relative to U, Online reduces the mean bound by 38.8%, from 0.05028 to 0.03078, while coverage changes from 95.07% to 89.59%, closer to the nominal 90% level in every scene. These results show less conservative bounds with near-target empirical coverage. However, Online has slightly higher mean ATE and RPE than U, so this improvement in bound tightness does not translate into better mean tracking accuracy.

Online calibration. We compare fixed calibration with the feedback mechanism in Sec. B.3.4. Fig. 7 shows that feedback raises mean coverage from 87.87% to 89.59%, bringing every scene closer to 90%. Coverage and mean bound are measured on reserved RGB pixels before map updates and averaged over 399 evaluated frames per scene. This comparison supports feedback as a mechanism for reducing empirical undercoverage as the map evolves. Table 10 also reports ATE and RPE improvements on six scenes each. Mean ATE decreases from 0.293 to 0.290 cm, while throughput remains approximately 1.06 FPS.

Trajectory visualization. Fig. 8 overlays ground-truth and estimated trajectories on the maps, using a shared viewpoint and two enlarged regions selected from the ground-truth trajectory. All trajectories retain their evaluated rigid alignment, without coordinate offsets or error amplification. The maps provide spatial context. The evaluation concerns

Table 6. **Coverage and bound size before and after calibration.** The nominal coverage is 90%. Mean bounds are in RGB-MAE units for RGB values in $[0, 1]$. Shaded rows are equal-scene means.

Dataset	Scene	Coverage (%)		Mean bound	
		QuantileGS-U	**QuantileGS-C**	QuantileGS-U	**QuantileGS-C**
LLFF	fern	57.29	89.57	0.0290	0.0953
	flower	73.49	93.48	0.0337	0.0713
	fortress	66.36	89.53	0.0254	0.0503
	horns	82.40	89.39	0.0408	0.0529
	leaves	65.55	89.96	0.0907	0.2319
	orchids	51.18	88.94	0.0457	0.1837
	room	78.21	91.39	0.0206	0.0373
	trex	81.97	84.59	0.0448	0.0490
	Average	69.56	89.61	0.0413	0.0965
Mip-NeRF 360	bicycle	75.84	94.91	0.0596	0.1345
	bonsai	87.15	91.40	0.0350	0.0431
	counter	85.30	91.59	0.0395	0.0545
	flowers	81.54	92.73	0.0957	0.1485
	garden	83.49	90.44	0.0489	0.0642
	kitchen	87.30	91.16	0.0396	0.0478
	room	83.62	90.82	0.0335	0.0474
	stump	64.19	91.48	0.0444	0.1351
	treehill	72.23	94.82	0.0656	0.1606
	Average	80.07	92.15	0.0513	0.0929
LF	africa	96.66	97.12	0.0696	0.0748
	basket	94.84	93.94	0.0816	0.0736
	statue	98.70	98.66	0.0638	0.0625
	torch	95.07	95.83	0.0584	0.0656
	Average	96.32	96.39	0.0684	0.0691

camera tracking rather than global reconstruction quality.

Table 7. **Compression, coverage, and bound size after pruning.** The 20% and 10% columns denote Gaussian retention levels. Counts are identical across the three compared methods. QuantileGS bounds for pruned models are transported from the original calibrated model. Coverage (%) and mean RGB-MAE bounds pool test pixels within each scene. Average weights scenes equally. All evaluated coverage-implication violation counts are zero.

Scene	Gaussian count			Coverage (%)			Mean bound		
	Original	20%	10%	Original	20%	10%	Original	20%	10%
bicycle	4 393 644	878 729	439 365	94.95	97.80	97.94	0.1350	0.1597	0.1630
bonsai	1 102 270	220 454	110 227	91.31	95.81	95.99	0.0425	0.0523	0.0545
counter	1 015 704	203 141	101 571	91.67	95.91	96.08	0.0554	0.0664	0.0690
flowers	2 712 997	542 600	271 300	92.79	96.38	96.79	0.1495	0.1755	0.1828
garden	3 309 450	661 890	330 945	90.42	95.39	95.67	0.0642	0.0834	0.0880
kitchen	1 423 911	284 783	142 392	91.26	95.76	96.33	0.0480	0.0594	0.0619
room	1 279 281	255 857	127 929	90.65	95.38	95.64	0.0474	0.0566	0.0582
stump	3 517 089	703 418	351 709	91.53	97.87	98.19	0.1354	0.1706	0.1750
treehill	3 204 008	640 802	320 401	94.68	97.64	97.84	0.1586	0.1885	0.1910
Average	2 439 817.1	487 963.8	243 982.1	92.14	96.44	96.72	0.0929	0.1125	0.1159

Table 8. **Effect of the calibration target.** Equal-scene means over nine unpruned Mip-NeRF 360 scenes (seed 0), using the same RGB renderer, fitted quantile fields, and test observations. C80/C90/C95 differ only in calibration target; U has no prescribed conformal target. The 90% setting is shaded.

Variant	Target (%)	Coverage (%)	Mean bound
U	–	80.01	0.0513
C80	80	82.71	0.0593
C90	90	92.14	0.0929
C95	95	96.65	0.1379

Table 9. **UQ and calibration in RGB-D SLAM.** Eight-scene means for seed 0. ATE and RPE are in centimeters. Coverage and mean RGB-MAE bounds are measured before reference-color updates. Darker shades indicate better tracking ranks.

Method	Tracking (cm)		Pre-update UQ	
	ATE ↓	RPE ↓	Cov. (%)	Mean bound
No-UQ	0.3427	0.01021	–	–
U	**0.2800**	**0.00443**	95.07	0.05028
Online	0.2898	0.00455	89.59	0.03078

Table 10. **Fixed and online SLAM calibration on Replica.** Both variants use seed 0; Online uses a step size of 0.01. ATE and translational RPE are RMSE values in centimeters. Bold marks the better value in each Fixed/Online pair. Darker shades indicate better ranks in the equal-scene Average row.

Scene	ATE ↓		RPE ↓		FPS ↑	
	Fixed	Online	Fixed	Online	Fixed	Online
room0	0.493	**0.483**	0.00586	**0.00577**	1.028	**1.051**
room1	0.280	**0.279**	0.00512	**0.00505**	1.008	**1.023**
room2	0.278	**0.275**	0.00405	**0.00399**	**1.025**	1.014
office0	0.372	**0.368**	0.00388	**0.00387**	1.103	**1.108**
office1	0.164	**0.158**	**0.00357**	0.00359	**1.096**	1.065
office2	**0.258**	0.263	**0.00436**	0.00441	1.035	**1.052**
office3	**0.270**	0.272	0.00450	**0.00446**	1.042	**1.051**
office4	0.227	**0.221**	0.00538	**0.00527**	**1.109**	1.097
Average	0.293	**0.290**	0.00459	**0.00455**	1.056	**1.058**

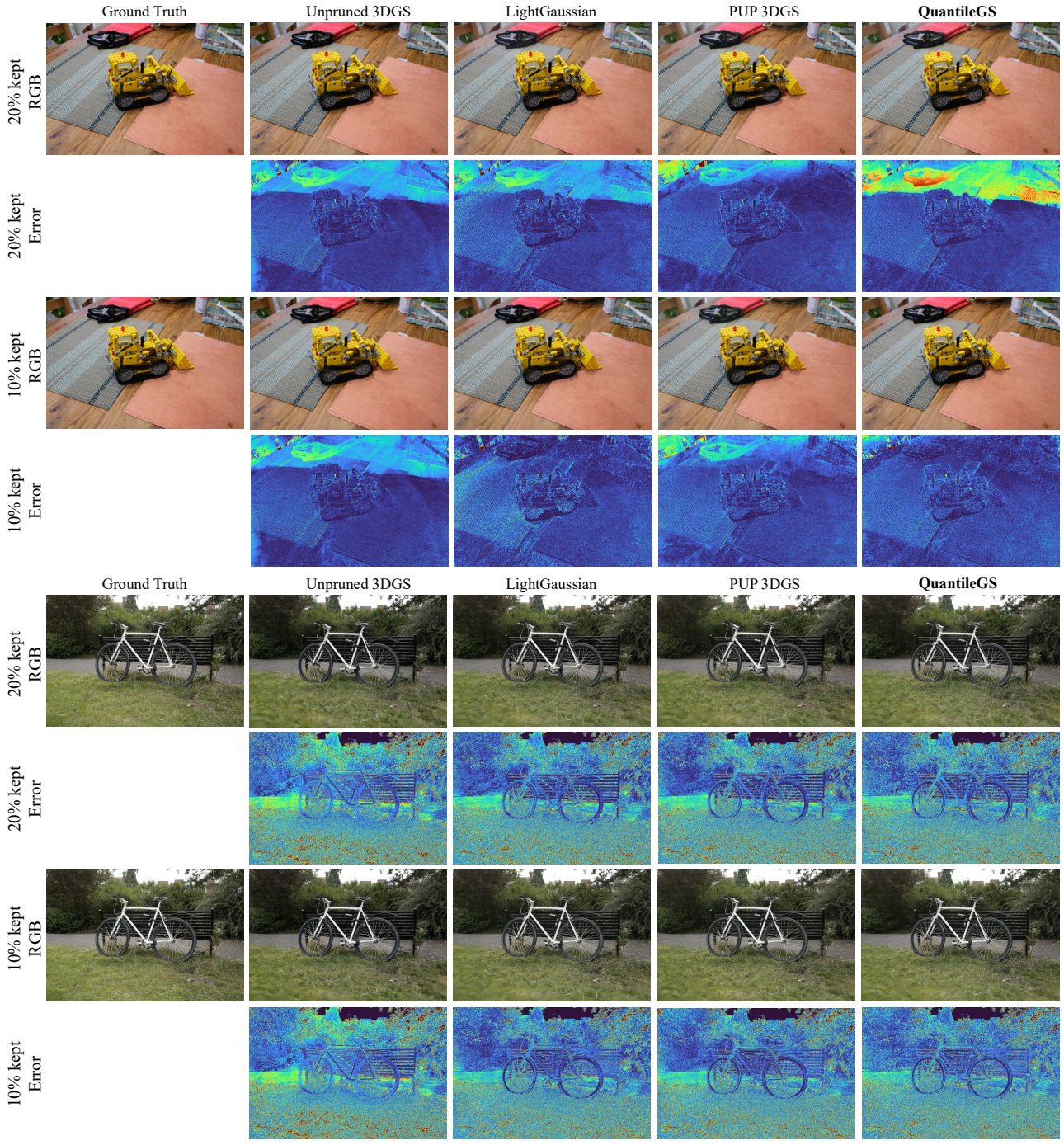

Figure 5. Pruning comparisons on Mip-NeRF 360 kitchen (top) and bicycle (bottom). Columns show ground truth, unpruned 3DGS, LightGaussian, PUP 3D-GS, and QuantileGS. Within each scene, the upper and lower row pairs show RGB renderings and RGB-MAE maps at 20% and 10% retention, respectively. All error maps share the $[0, 0.20]$ color scale, with errors above 0.20 shown at the maximum color.

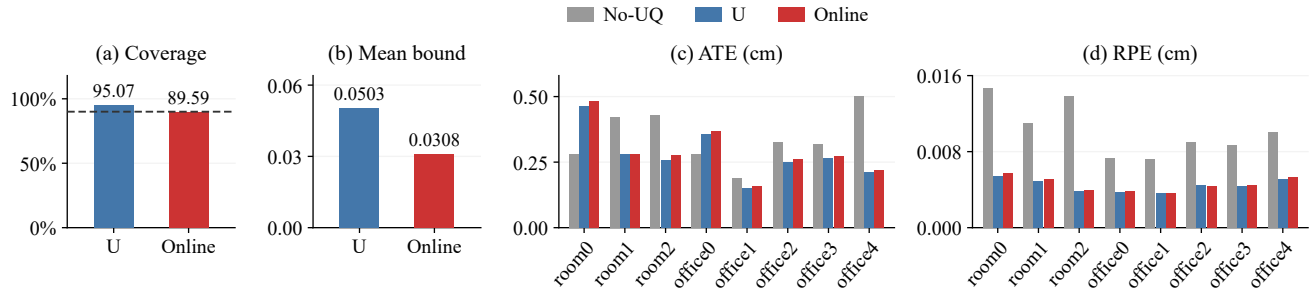

Figure 6. Uncertainty and calibration on eight Replica sequences. (a,b) Mean pre-update RGB coverage and RGB-MAE bounds. The dashed line marks 90%. (c,d) Per-scene ATE and translational RPE in centimeters. Online reuses the previously evaluated runs.

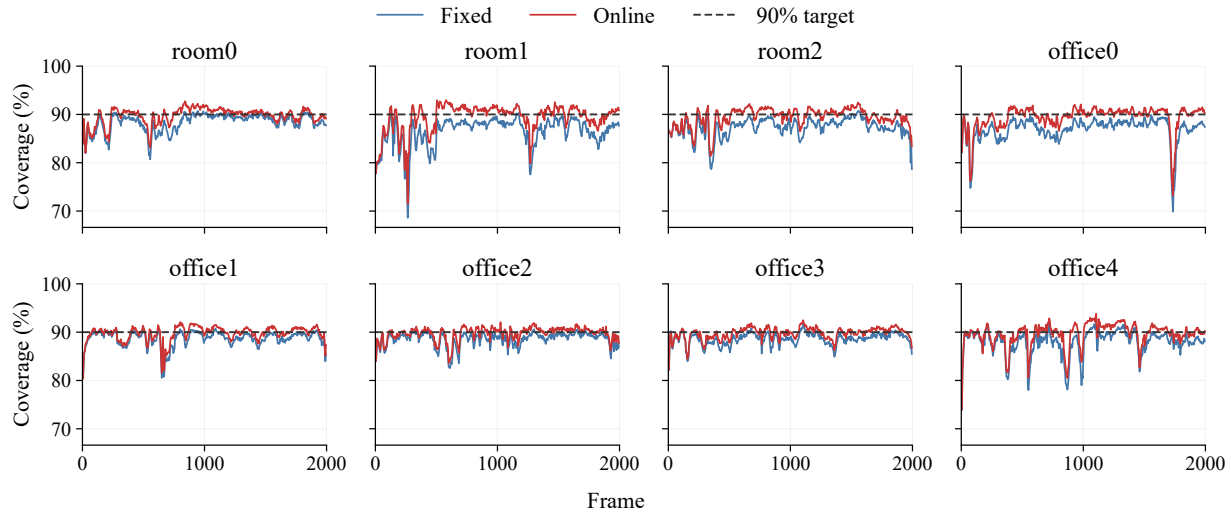

Figure 7. Pre-update RGB holdout coverage on eight Replica sequences. Blue and red denote fixed and online calibration. The dashed line marks 90%. Each curve connects 399 evaluation frames without smoothing.

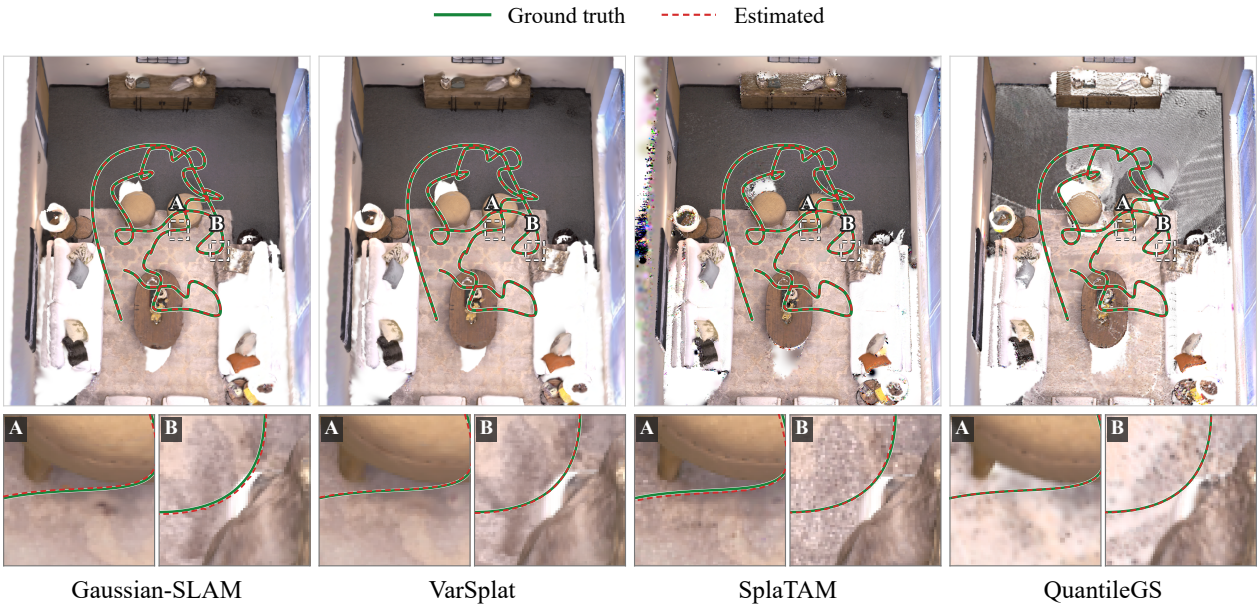

Figure 8. Trajectory comparison on Replica room0. Each column shows a method's map and shared enlarged regions A and B. Ground truth is green and estimated trajectories are red. Maps and trajectories retain their evaluated rigid alignment.